AS · Unit 2 · WCH12 Energetics, Group Chemistry, Halogenoalkanes and Alcohols

Energetics

Every chemical reaction takes in or gives out energy, because bonds have to be broken and new ones made. This topic shows you how to define, measure and calculate these enthalpy changes: with a thermometer and a polystyrene cup, with Hess cycles that find values nobody can measure directly, and with bond enthalpies.

Enthalpy changes

In energetics we split the world into two parts. The **system** is the chemicals that are reacting. The **surroundings** are everything else: the water they are dissolved in, the beaker, the thermometer, the air. When a reaction happens, energy moves between the two, and the total amount of energy stays the same.

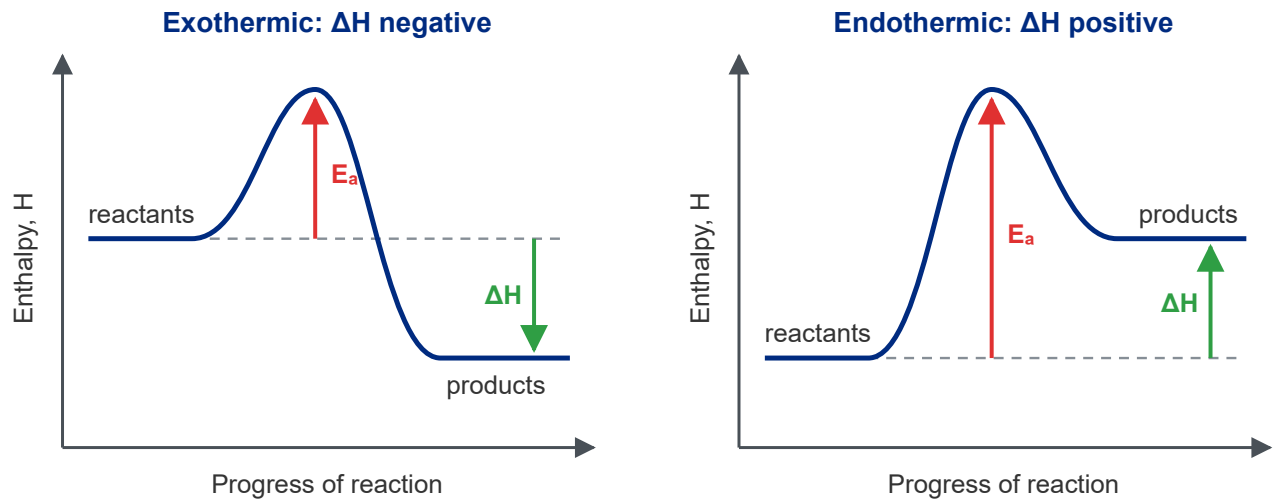
The heat content of a system is called its **enthalpy**, H. Nobody can measure the enthalpy of a substance on its own, but we can measure how much it changes. The **enthalpy change**, ΔH, is the heat energy transferred between the system and the surroundings when a reaction takes place at **constant pressure**. Most reactions in a lab happen in open containers, at the pressure of the atmosphere, so this is the quantity we actually measure.

- **Exothermic:** the system gives heat energy out to the surroundings, so the surroundings warm up. The products have less enthalpy than the reactants, so ΔH is **negative**. Combustion, neutralisation and respiration are all exothermic.
- **Endothermic:** the system takes heat energy in from the surroundings, so the surroundings cool down (or the reaction needs continuous heating). The products have more enthalpy than the reactants, so ΔH is **positive**. Thermal decomposition, such as $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$, and photosynthesis are endothermic.

Why do reactions transfer energy at all? Breaking a bond always **takes in** energy, because you have to pull the atoms apart against the attraction between the bonding electrons and the two nuclei. Making a bond always **gives out** energy. If the bonds made in the products are stronger overall than the bonds broken in the reactants, more energy comes out than goes in and the reaction is exothermic. If the bonds broken are stronger overall, the reaction is endothermic.

Enthalpy level diagrams and reaction profiles

An enthalpy level diagram puts the reactants and products at heights that show their enthalpy. The ΔH arrow always starts at the reactants and points to the products, so it points **down** for an exothermic reaction and **up** for an endothermic one. A reaction profile adds the path between them. Even an exothermic reaction has to climb a hump first: some bonds must start to break before new ones can form. The height of that hump above the reactants is the **activation energy**, E_a, the minimum energy that colliding particles need for a reaction to happen.



Reaction profiles. The activation energy is measured from the reactants up to the top of the hump. ΔH is the difference between the reactants and the products, whatever happens in between.

EXAM TIP

When you draw an enthalpy level diagram, label the vertical axis "enthalpy", write the formulas of the reactants and products on their levels, and draw the ΔH arrow **from reactants to products** with a single arrowhead. An exam answer that has the arrow pointing the wrong way loses the mark even if the levels are right.

Standard conditions

The size of an enthalpy change depends on the temperature, the pressure, the concentrations and the physical states of the substances. Water vapour condensing to liquid water gives out extra energy, for example, so burning methane to make $H_2O(l)$ gives a bigger ΔH than burning it to make $H_2O(g)$. To compare values fairly, chemists quote them under **standard conditions** and mark them with the symbol $\ominus$, as in $\Delta H^{\ominus}$.

- **Pressure:** 100 kPa.
- **Temperature:** a stated temperature, normally 298 K (25 °C).
- **Solutions:** concentration 1.00 mol dm^{-3}.
- **Standard states:** every substance is in the physical state it normally has at 100 kPa and 298 K. Water is $H_2O(l)$, oxygen is $O_2(g)$, bromine is $Br_2(l)$, iodine is $I_2(s)$, carbon is C(s) in the form of graphite.

ΔH values are quoted in kJ mol^{-1}. The "per mole" refers to the equation as written: if you double every number in the equation, you double ΔH.

Standard enthalpy change definitions

You need to know five definitions word for word in meaning. Each one fixes three things: **what reacts**, **how much** (usually exactly one mole of something), and the **conditions and states**.

DEFINITION

Standard enthalpy change of reaction, $\Delta_r H^{\ominus}$

The enthalpy change when the amounts of reactants shown in the equation react together under standard conditions, with all substances in their standard states.

DEFINITION

Standard enthalpy change of formation, $\Delta_f H^{\ominus}$

The enthalpy change when **one mole** of a compound is formed from its **elements** in their standard states, under standard conditions.

The equation must make exactly one mole of the compound, which often means fractions in front of the elements:

$$2C(s) + 3H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow C_2H_5OH(l) \quad \Delta_f H^{\ominus} = -277 \text{ kJ mol}^{-1}$$

$$Na(s) + \tfrac{1}{2}Cl_2(g) \rightarrow NaCl(s) \quad \Delta_f H^{\ominus} = -411 \text{ kJ mol}^{-1}$$

The standard enthalpy of formation of an **element in its standard state is zero**. Forming $O_2(g)$ from $O_2(g)$ is no change at all. This is not true of other forms of the element: $O_3(g)$ or $Br_2(g)$ have non-zero values.

DEFINITION

Standard enthalpy change of combustion, $\Delta_c H^{\ominus}$

The enthalpy change when **one mole** of a substance is burned **completely in excess oxygen** under standard conditions, with all reactants and products in their standard states.

$$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l) \quad \Delta_c H^{\ominus} = -1367 \text{ kJ mol}^{-1}$$

Combustion is always exothermic. Complete combustion of a compound of carbon and hydrogen gives only CO_2 and H_2O. With too little oxygen you get incomplete combustion, making carbon monoxide and soot (carbon). Less energy is released, because the carbon has not been fully oxidised. Notice also that the combustion of hydrogen, $H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(l)$, is the same reaction as the formation of water, so $\Delta_c H^{\ominus}(H_2) = \Delta_f H^{\ominus}(H_2O(l)) = -286$ kJ mol^{-1}. In the same way, $\Delta_c H^{\ominus}$(C, graphite) $= \Delta_f H^{\ominus}(CO_2) = -394$ kJ mol^{-1}.

DEFINITION

Standard enthalpy change of neutralisation, $\Delta_{neut}H^{\ominus}$

The enthalpy change when an acid and an alkali react together in solution to form **one mole of water**, under standard conditions.

For any strong acid with any strong alkali the value is about −57 kJ mol^{-1}. The reason is that strong acids and alkalis are fully ionised, so the only reaction that actually happens is the same every time:

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

The other ions (Na^+, Cl^-, K^+, NO_3^- and so on) are spectators. A weak acid such as ethanoic acid gives a slightly less exothermic value, because some energy is used up in ionising the acid as it reacts. The definition is per mole of **water**: $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$ makes two moles of water, so the enthalpy change for that equation is about twice $\Delta_{neut}H^{\ominus}$.

DEFINITION

Standard enthalpy change of atomisation, $\Delta_{at}H^{\ominus}$

The enthalpy change when **one mole of gaseous atoms** is formed from an element in its standard state, under standard conditions.

$$Na(s) \rightarrow Na(g) \quad \Delta_{at}H^{\ominus} = +107 \text{ kJ mol}^{-1}$$
$$\tfrac{1}{2}Cl_2(g) \rightarrow Cl(g) \quad \Delta_{at}H^{\ominus} = +122 \text{ kJ mol}^{-1}$$

Atomisation is always endothermic, because bonds (metallic bonds in sodium, the covalent Cl–Cl bond in chlorine) have to be broken. For chlorine, the definition is per mole of Cl *atoms*, so it is half the energy needed to break one mole of Cl–Cl bonds.

Enthalpy change	Fixed amount	Sign
reaction, $\Delta_rH^{\ominus}$	the molar amounts in the equation as written	either
formation, $\Delta_fH^{\ominus}$	1 mol of compound made from its elements	usually negative
combustion, $\Delta_cH^{\ominus}$	1 mol of substance burned completely	always negative
neutralisation, $\Delta_{neut}H^{\ominus}$	1 mol of water formed	always negative
atomisation, $\Delta_{at}H^{\ominus}$	1 mol of gaseous atoms formed	always positive

COMMON MISTAKE

Two common slips in definitions: forgetting the state symbols, and getting the "one mole" wrong. $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$ is *not* the formation equation of water, because it makes two moles. And $H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow$

$H_2O(g)$ is not either, because water's standard state is liquid.

Measuring enthalpy changes: calorimetry

You can't measure the heat a reaction gives out directly. What you can measure is its effect: the temperature change of a known mass of water or solution that absorbs the heat. The energy needed to warm a substance depends on its mass, on how far its temperature changes, and on how much energy that substance needs per gram per degree, its **specific heat capacity**, c.

$$Q = mc\,\Delta T$$

- **Q:** heat energy transferred, in J.
- **m:** the mass of the water or solution whose temperature changes, in g. Not the mass of the reactants, and not the mass of any solid added.
- **c:** specific heat capacity. For water and dilute solutions, use 4.18 J g^{-1} K^{-1}.
- **ΔT:** temperature change. A change of 1 K is the same size as a change of 1 °C, so either scale works for ΔT.

Q is the energy for the amounts you actually used. To turn it into ΔH you divide by the number of moles of the substance that sets the amount of reaction, then convert J into kJ and add the sign: a temperature **rise** means the reaction gave out heat, so ΔH is **negative**.

$$\Delta H = \frac{-Q}{n}$$

Reactions in solution

For neutralisation, displacement, dissolving and similar reactions, the solution itself absorbs the heat. The reaction is done in an expanded polystyrene cup, which is a good insulator and has a very small heat capacity, so almost all the energy stays in the solution.

- Stand the polystyrene cup in a beaker for support and extra insulation, and use a lid with a hole for the thermometer.
- Measure the solution into the cup with a pipette or burette. Weigh any solid reactant accurately, preferably by weighing the container before and after adding the solid.
- Record the temperature every minute for 3 minutes to get a steady starting value. If two solutions are mixed, measure both and use the average.
- At the fourth minute add the second reactant, stir, and keep recording every half-minute or minute for several more minutes. Stirring makes sure all of the solution is at the same temperature.
- Plot temperature against time and **extrapolate** the cooling line back to the moment of mixing to find ΔT.

Why extrapolate? While the reaction is still giving out heat, the warm solution is already losing some to the room. The highest temperature you read is therefore lower than it would have been if all the heat had been released instantly. The steady cooling afterwards shows how fast heat is being lost; following that line back to the time of

mixing estimates the temperature the solution would have reached with no losses. The slower the reaction, the more this matters.

Worked example: a displacement reaction

1. 50.0 cm^3 of 0.500 mol dm^{-3} copper(II) sulfate solution is placed in a polystyrene cup and an excess of zinc powder is added. The extrapolated temperature rise is 24.5 °C. Calculate the enthalpy change for $Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$.

2. Heat absorbed by the solution. The mass is that of 50.0 cm^3 of solution, taken as 50.0 g:

$$Q = mc\Delta T = 50.0 \times 4.18 \times 24.5 = 5120.5 \text{ J}$$

The zinc is a solid that does not get counted in m. We also assume the solution has the density and heat capacity of water.

3. Moles of the reactant that is not in excess, copper(II) sulfate:

$$n = cV = 0.500 \times \frac{50.0}{1000} = 0.0250 \text{ mol}$$

4. Divide, convert to kJ, and give it a negative sign because the temperature rose:

$$\Delta_r H = \frac{-5120.5}{0.0250} = -204\,820 \text{ J mol}^{-1} = -205 \text{ kJ mol}^{-1}$$

The data book value is −217 kJ mol^{-1}. The experiment gives a smaller magnitude because some heat still escaped and some warmed the cup and thermometer.

Worked example: neutralisation

1. 50.0 cm^3 of 1.00 mol dm^{-3} hydrochloric acid and 50.0 cm^3 of 1.00 mol dm^{-3} sodium hydroxide solution, both at 19.8 °C, are mixed. The extrapolated maximum temperature is 26.4 °C. Calculate the enthalpy change of neutralisation.

2. Both solutions end up warmer, so the mass is the total volume, 100.0 cm^3 = 100.0 g. ΔT = 26.4 − 19.8 = 6.6 K.

$$Q = 100.0 \times 4.18 \times 6.6 = 2758.8 \text{ J}$$

3. The equation is HCl + NaOH → NaCl + H_2O, so the moles of water formed equal the moles of acid (and of alkali):

$$n(\mathrm{H_2O}) = 1.00 \times \frac{50.0}{1000} = 0.0500 \text{ mol}$$

4. $$\Delta_{\text{neut}}H = \frac{-2758.8}{0.0500} = -55\,176 \text{ J mol}^{-1} = -55.2 \text{ kJ mol}^{-1}$$

Always finish with a sign, units and a sensible number of significant figures (normally 3, or fewer if the data have fewer).

COMMON MISTAKE

The most common errors in these calculations: using only one solution's volume for *m* when two solutions are mixed, adding the mass of a solid to *m*, forgetting to divide by 1000 to get kJ, and leaving off the minus sign for an exothermic reaction.

Sources of error and assumptions

- **Heat loss** (or gain) to the surroundings, the biggest error. It makes the measured ΔH less exothermic (or less endothermic) than the true value.
- **Heat capacity of the apparatus** is ignored: the cup and thermometer absorb a little energy too.
- **The solution is treated as water**: density 1.00 g cm^{-3} and c = 4.18 J g^{-1} K^{-1}.
- **The reaction may be incomplete or slow**, or a solid may not fully dissolve.
- **Uncertainty in ΔT:** a thermometer reading has an uncertainty of half a division, and ΔT uses two readings. A larger temperature change (more concentrated solutions, or more solid) makes the *percentage* uncertainty in ΔT smaller.

Enthalpy change of combustion

To measure $\Delta_c H$ of a liquid fuel such as an alcohol, burn it in a spirit burner under a copper can containing a known mass of water. Weigh the burner (with its cap on, to stop the fuel evaporating) before and after, measure the water's temperature rise, and use Q = mcΔT with the mass of the **water**.

Worked example: combustion of ethanol

1. A spirit burner containing ethanol, C_2H_5OH, loses 0.920 g while heating 200 g of water from 19.5 °C to 37.0 °C. Calculate $\Delta_c H$ for ethanol.

2 Energy absorbed by the water (ΔT = 17.5 K):

$$Q = 200 \times 4.18 \times 17.5 = 14\,630 \text{ J}$$

m is the mass of water in the can, not the mass of fuel burned.

3 Moles of ethanol burned, using M_r = 2(12.0) + 6(1.0) + 16.0 = 46.0:

$$n = \frac{0.920}{46.0} = 0.0200 \text{ mol}$$

4

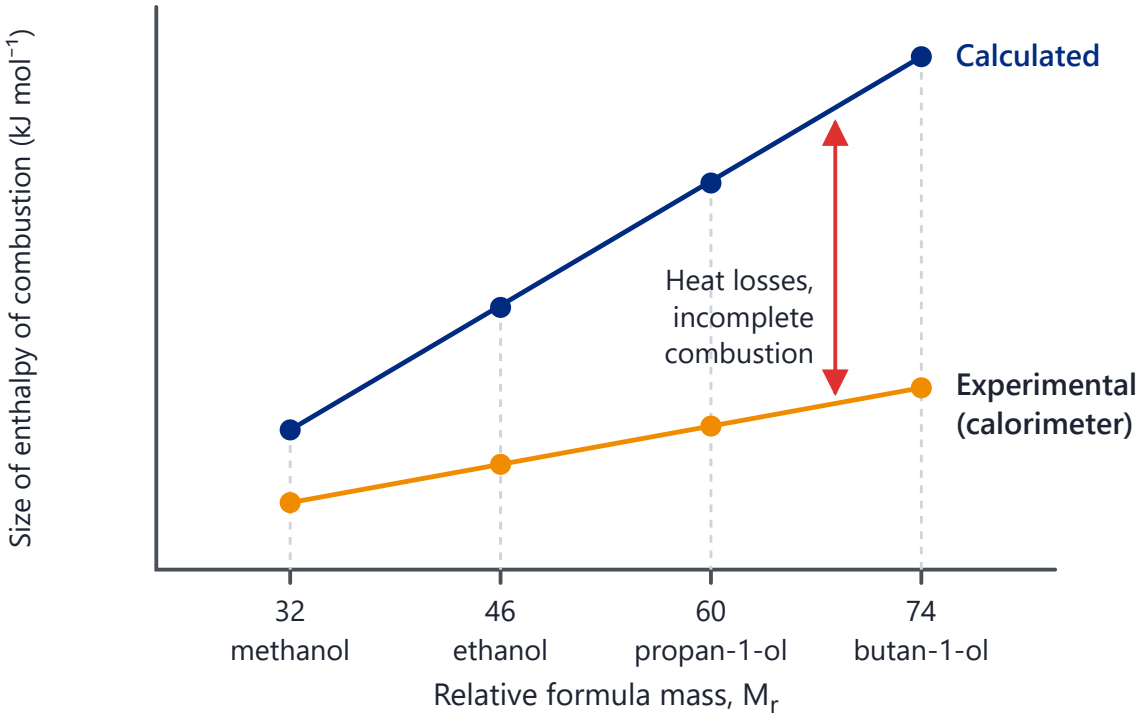
$$\Delta_c H = \frac{-14\,630}{0.0200} = -731\,500 \text{ J mol}^{-1} = -732 \text{ kJ mol}^{-1}$$

The data book value is −1367 kJ mol^{-1}, so only about half of the energy released reached the water.

Combustion calorimetry is much less accurate than the polystyrene-cup method. Most of the error comes from:

- **Heat loss:** hot gases from the flame escape around the can and warm the air instead of the water. A draught shield, a lid on the can and a short gap between flame and can all help.
- **Incomplete combustion:** soot on the bottom of the can shows that some carbon was not burned to CO_2, so less energy was released.
- **Evaporation** of the fuel from the wick between weighings, making the mass "burned" seem too large.
- **The heat capacity of the can** is ignored.
- **Non-standard conditions:** the water formed escapes as steam, not as liquid, so the energy it would release on condensing is lost.

Going up a homologous series, each extra CH_2 group means breaking one more C–C bond, two more C–H bonds and 1½ more O=O bonds, and making two more C=O bonds and two more O–H bonds. The extra amount of energy is the same each time, so $\Delta_c H$ becomes more negative by a roughly constant amount, about 650 kJ mol^{-1} per CH_2: methanol −726, ethanol −1367, propan-1-ol −2021, butan-1-ol −2676 kJ mol^{-1}.



Sketch graph (not to scale) of the size of the enthalpy of combustion against M_r for the alcohols. Both sets of values rise steadily with each CH_2 group, but the calorimeter results are far smaller because of heat losses and incomplete combustion.

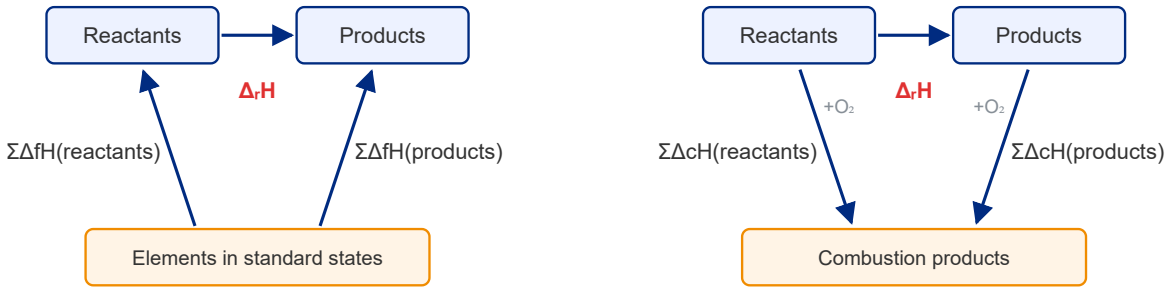
Hess's law

KEY IDEA

Hess's law: the total enthalpy change for a reaction is independent of the route taken, as long as the starting and finishing conditions are the same.

Hess's law follows from the conservation of energy. Imagine two routes from A to B with different enthalpy changes. You could go from A to B by the more exothermic route and come back by the reverse of the other, ending where you started with energy left over. That would create energy from nothing, which never happens.

The law is useful because many enthalpy changes can't be measured directly. You can't make carbon and hydrogen react to give methane in a calorimeter, and you can't heat calcium carbonate while measuring a temperature change. Instead, you build an **enthalpy cycle**: the direct route from reactants to products, plus an indirect route through some other substances whose enthalpy changes are known. The rule for going round a cycle is simple: **follow an arrow and you use its ΔH; go against an arrow and you change its sign**.



Left: a cycle using enthalpies of formation, with arrows pointing up out of the elements. Right: a cycle using enthalpies of combustion, with arrows pointing down into the combustion products (CO_2 and H_2O).

Using enthalpies of formation

Every compound can be made from its elements, so the elements make a convenient bottom corner for the cycle. Going from reactants to products the indirect way means first going backwards down the left arrow (breaking the reactants back into elements, so the sign changes) and then up the right arrow (forming the products). That gives:

$$\Delta_r H = \sum \Delta_f H(\text{products}) - \sum \Delta_f H(\text{reactants})$$

Worked example: reducing iron(III) oxide

1. Calculate $\Delta_r H$ for $Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(s) + 3CO_2(g)$. Data, $\Delta_f H$ / kJ mol^{-1}: $Fe_2O_3(s)$ −824.2, $CO(g)$ −110.5, $CO_2(g)$ −393.5.

2. Sum for the products. Iron is an element in its standard state, so its value is zero. Multiply each value by the number of moles in the equation:

$$\sum \Delta_f H(\text{products}) = 2(0) + 3(-393.5) = -1180.5 \text{ kJ mol}^{-1}$$

3. Sum for the reactants:

$$\sum \Delta_f H(\text{reactants}) = (-824.2) + 3(-110.5) = -1155.7 \text{ kJ mol}^{-1}$$

4. Products minus reactants:

$$\Delta_r H = -1180.5 - (-1155.7) = -24.8 \text{ kJ mol}^{-1}$$

Watch the double negative: subtracting −1155.7 is the same as adding 1155.7.

Using enthalpies of combustion

Enthalpies of combustion are easy to measure for anything that burns, and every organic compound burns to the same products, CO_2 and H_2O. So the combustion products make the bottom corner. This time the indirect route goes down the left arrow (burning the reactants) and then backwards up the right arrow (un-burning the products):

$$\Delta_r H = \sum \Delta_c H(\text{reactants}) - \sum \Delta_c H(\text{products})$$

COMMON MISTAKE

The two formulas have the subtraction the opposite way round. Don't learn them as a rhyme: draw the cycle and follow the arrows, and you can't get the order wrong.

Worked example: the enthalpy of formation of propane

1. Calculate $\Delta_f H$ of propane, $3C(s) + 4H_2(g) \rightarrow C_3H_8(g)$, from $\Delta_c H$ / kJ mol^{-1}: C(s) −393.5, $H_2(g)$ −285.8, $C_3H_8(g)$ −2219.2.

   Carbon and hydrogen won't combine to give just propane, so this can't be measured directly. All three substances burn easily, though.

2. Burn the reactants (down the left arrow):

$$\sum \Delta_c H(\text{reactants}) = 3(-393.5) + 4(-285.8) = -2323.7 \text{ kJ mol}^{-1}$$

3. The product: one mole of propane, $\sum \Delta_c H(\text{products}) = -2219.2 \text{ kJ mol}^{-1}$.

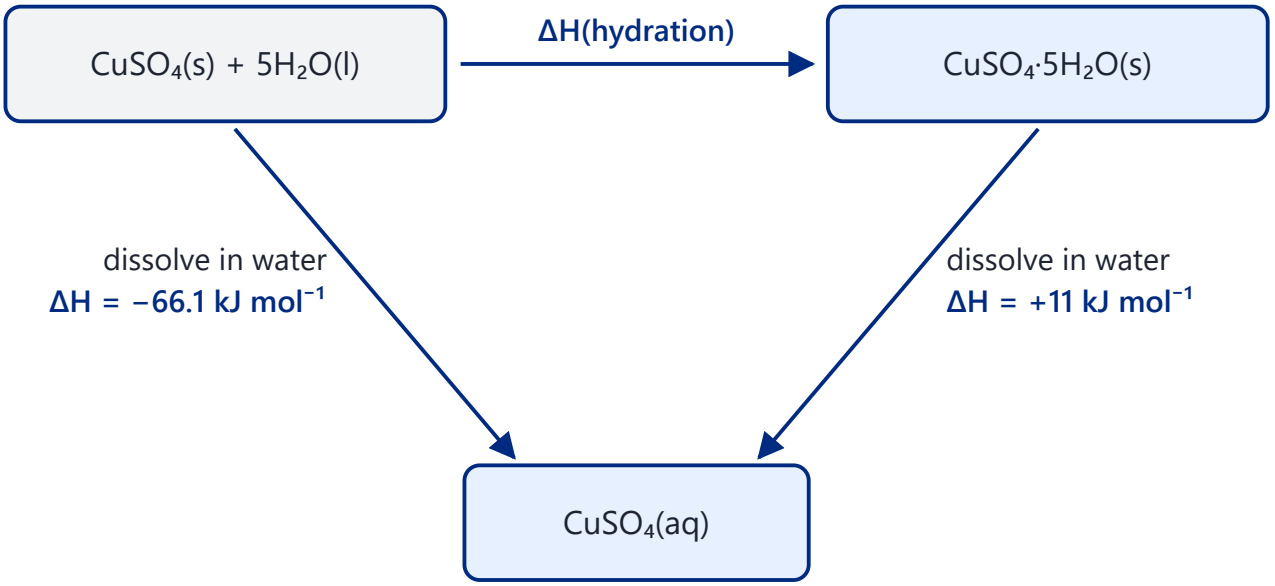
4. Reactants minus products:

$$\Delta_f H(C_3H_8) = -2323.7 - (-2219.2) = -104.5 \text{ kJ mol}^{-1}$$

Finding enthalpy changes that can't be measured directly

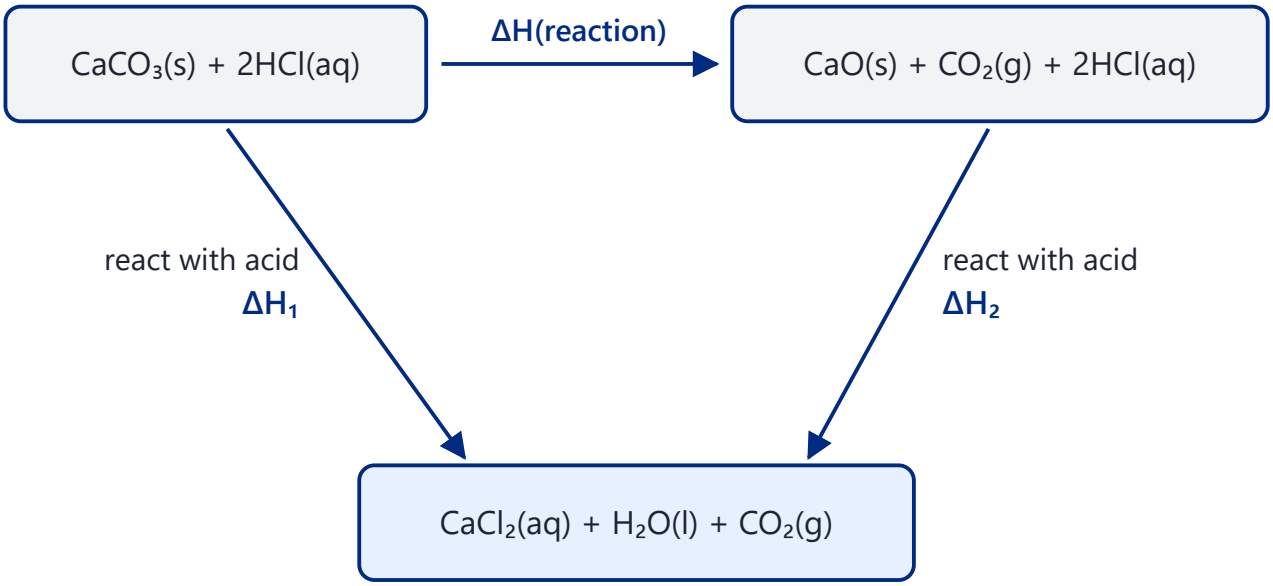
Some reactions are impossible to carry out cleanly in a calorimeter, but you can reach the same products by two reactions that *can* be measured.

Hydrating copper(II) sulfate. White anhydrous copper(II) sulfate turns into blue hydrated crystals when it takes up water: $CuSO_4(s) + 5H_2O(l) \rightarrow CuSO_4{\cdot}5H_2O(s)$. You can't add exactly five moles of water and measure the temperature of a solid. Instead, dissolve each solid separately in plenty of water. Both give the same solution, $CuSO_4(aq)$, so they share a corner of the cycle.



Both salts dissolve to give $CuSO_4(aq)$. Following the arrows, ΔH(hydration) + 11 = −66.1, so ΔH(hydration) = −77.1 kJ mol^{-1}.

Decomposing calcium carbonate. $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ needs strong heating, so you can't measure its heat change with a thermometer. Both calcium carbonate and calcium oxide react with hydrochloric acid to give the same products, and those two reactions can be done in a polystyrene cup. ΔH_1 and ΔH_2 are measured, and $\Delta H = \Delta H_1 - \Delta H_2$. (The CO_2 from the decomposition simply ends up in the bottom box alongside the CO_2 from the acid reaction.)



Calcium carbonate and calcium oxide both react with hydrochloric acid to give the same products. Going round the cycle, $\Delta H(\text{reaction}) + \Delta H_2 = \Delta H_1$.

CORE PRACTICAL

Core Practical 2: determine the enthalpy change of a reaction using Hess's law

A typical version finds the enthalpy change for hydrating anhydrous copper(II) sulfate.

- Pipette 50.0 cm^3 of distilled water into a polystyrene cup standing in a beaker. Record its temperature every minute for three minutes.
- Weigh about 4 g of anhydrous copper(II) sulfate accurately in a dry weighing bottle. At the fourth minute tip it in, stir, and record the temperature every half-minute until about the fifteenth minute. Reweigh the empty bottle to get the exact mass added.
- Repeat with about 6.2 g of hydrated copper(II) sulfate, $CuSO_4 \cdot 5H_2O$, in a fresh 50.0 cm^3 of water. This time the temperature falls.
- Plot both temperature–time graphs, extrapolate to the time of mixing to find ΔT, and work out ΔH for each dissolving per mole of salt.
- **Safety:** copper(II) sulfate is harmful if swallowed and irritates the eyes, so wear eye protection.

Another version uses potassium hydrogencarbonate and potassium carbonate, each reacted with excess hydrochloric acid, to find the enthalpy change of $2KHCO_3(s) \rightarrow K_2CO_3(s) + CO_2(g) + H_2O(l)$. You can run both of those reactions, and both copper sulfate experiments, in the calorimetry simulation above.

Worked example: Core Practical 2 results

1. Experiment 1: 3.99 g of anhydrous $CuSO_4$ (M_r = 159.6) dissolved in 50.0 g of water. Extrapolated ΔT = +7.6 K.

$$Q = 50.0 \times 4.18 \times 7.6 = 1588.4\ \text{J}, \qquad n = \frac{3.99}{159.6} = 0.0250\ \text{mol}$$

$$\Delta H_1 = \frac{-1588.4}{0.0250} = -63.5\ \text{kJ mol}^{-1}$$

2. Experiment 2: 6.24 g of $CuSO_4 \cdot 5H_2O$ (M_r = 249.6) dissolved in 50.0 g of water. Extrapolated ΔT = −1.3 K, so heat was taken in from the water.

$$Q = 50.0 \times 4.18 \times 1.3 = 271.7\ \text{J}, \qquad n = \frac{6.24}{249.6} = 0.0250\ \text{mol}$$

$$\Delta H_2 = \frac{+271.7}{0.0250} = +10.9\ \text{kJ mol}^{-1}$$

A temperature fall means an endothermic process, so this ΔH is positive.

3. Go round the cycle. Route 1: anhydrous salt + water → solution, ΔH_1. Route 2: anhydrous salt + water → hydrated salt (ΔH, unknown), then hydrated salt → solution (ΔH_2).

$$\Delta H + \Delta H_2 = \Delta H_1 \quad \Rightarrow \quad \Delta H = -63.5 - (+10.9) = -74.4\ \text{kJ mol}^{-1}$$

The data book value is about −78 kJ mol^{-1}. The biggest error is in ΔH_2: a fall of only 1.3 K has a large percentage uncertainty.

Bond enthalpies

DEFINITION

Bond enthalpy

The enthalpy change needed to break one mole of a particular covalent bond, with the molecules and the atoms formed all in the **gaseous** state. For example, $H_2(g) \rightarrow 2H(g)$, ΔH = +436 kJ mol^{-1}.

DEFINITION

Mean bond enthalpy

The enthalpy change needed to break one mole of a given type of bond, **averaged over many different**

compounds, with all species gaseous.

Bond enthalpies are always **positive**, because breaking a bond always needs energy. The strength of a bond depends a little on its surroundings. Breaking the four C–H bonds in methane one after another needs four different amounts of energy, because each time the remaining fragment is different. The C–H bond in ethanol is slightly different again. Data books therefore list a **mean** value for each type of bond, such as 413 kJ mol^{-1} for C–H, which is close to, but not exactly, the value in any one molecule.

Bond	Mean bond enthalpy / kJ mol^{-1}	Bond	Mean bond enthalpy / kJ mol^{-1}
H–H	436	O=O	498
C–H	413	O–H	464
C–C	347	C=O (in CO_2)	805
C=C	612	C–O	358
N≡N	945	N–H	391
Cl–Cl	243	H–Cl	432
Br–Br	193	H–Br	366
I–I	151	H–I	298

To estimate ΔH for a reaction between gases, imagine it happening in two stages: break every bond in the reactants to make separate gaseous atoms (energy in), then join the atoms up into the products (energy out). By Hess's law that is the same as the real reaction:

$$\Delta H = \sum(\text{bond enthalpies of bonds broken}) - \sum(\text{bond enthalpies of bonds made})$$

Worked example: burning methane, using mean bond enthalpies

1. Estimate ΔH for $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$. Draw out the molecules to count the bonds: methane has four C–H bonds, each O_2 has one O=O, CO_2 is O=C=O, and each water has two O–H bonds.

2. Bonds broken (endothermic):

$$4(\text{C–H}) + 2(\text{O=O}) = 4(413) + 2(498) = 1652 + 996 = +2648 \text{ kJ mol}^{-1}$$

3. Bonds made (exothermic):

$$2(\text{C=O}) + 4(\text{O–H}) = 2(805) + 4(464) = 1610 + 1856 = 3466 \text{ kJ mol}^{-1}$$

4. $$\Delta H = 2648 - 3466 = -818 \text{ kJ mol}^{-1}$$

Using enthalpies of formation for the same reaction (with water as a gas) gives −802 kJ mol^{-1}. The difference comes from using mean values instead of the actual bond enthalpies in these molecules.

Values from mean bond enthalpies are **less accurate** than values from enthalpies of formation or combustion, for two reasons. The mean values are averages, not the true values for the bonds in these particular molecules. And the method only works for gases: if a substance is a liquid or solid in its standard state, extra energy is involved in separating its molecules, which bond enthalpies leave out. That is why the methane example uses $H_2O(g)$; the standard $\Delta_c H$, which makes $H_2O(l)$, is −890 kJ mol^{-1}.

Working out a bond enthalpy from other data

You can turn the method round and use a known ΔH to find a bond enthalpy. If you need a substance as gaseous atoms and it is not a simple diatomic gas, use its enthalpy of atomisation instead of a bond enthalpy: carbon, C(s) → C(g), needs +717 kJ mol^{-1}.

Worked example: the N–H bond enthalpy in ammonia

1. $\Delta_f H(NH_3(g))$ = −46.1 kJ mol^{-1}. Use E(N≡N) = 945 and E(H–H) = 436 kJ mol^{-1} to find the N–H bond enthalpy. The formation equation is $\frac{1}{2}N_2(g) + 1\frac{1}{2}H_2(g) \rightarrow NH_3(g)$.

2. Bonds broken: half a mole of N≡N and one and a half moles of H–H:

$$\tfrac{1}{2}(945) + 1\tfrac{1}{2}(436) = 472.5 + 654 = 1126.5 \text{ kJ mol}^{-1}$$

3. Bonds made: three moles of N–H. Write the equation:

$$-46.1 = 1126.5 - 3E(\text{N–H})$$

4. Rearrange:

$$3E(\text{N–H}) = 1126.5 + 46.1 = 1172.6 \quad \Rightarrow \quad E(\text{N–H}) = 391 \text{ kJ mol}^{-1}$$

Bond enthalpies also give a feel for how reactive a substance will be. A reaction usually has to start by breaking a bond, so weak bonds (such as the I–I bond, 151 kJ mol^{-1}) break more easily and lead to faster reactions at room temperature than very strong bonds like N≡N (945 kJ mol^{-1}). That is one reason nitrogen is so unreactive.

Quick checks

QUICK CHECK

Which equation represents the standard enthalpy change of formation of ethanol?

- $C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$
- $2C(s) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$
- $4C(s) + 6H_2(g) + O_2(g) \rightarrow 2C_2H_5OH(l)$
- $2C(g) + 6H(g) + O(g) \rightarrow C_2H_5OH(l)$

Formation means one mole of the compound made from its elements in their standard states. The third option makes two moles, and the last one starts from gaseous atoms rather than the elements' standard states (C(s), $H_2(g)$, $O_2(g)$). The first is the combustion equation.

QUICK CHECK

25.0 cm^3 of 2.00 mol dm^{-3} HCl is mixed with 25.0 cm^3 of 2.00 mol dm^{-3} NaOH. The temperature rises by 13.0 °C. What is the enthalpy change of neutralisation?

- -27.2 kJ mol^{-1}
- -109 kJ mol^{-1}
- $+54.3$ kJ mol^{-1}
- -54.3 kJ mol^{-1}

The mass is the total of both solutions, 50.0 g, so Q = 50.0 × 4.18 × 13.0 = 2717 J. Moles of water = 2.00 × 0.0250 = 0.0500 mol, so ΔH = −2717 ÷ 0.0500 = −54 340 J mol^{-1} = −54.3 kJ mol^{-1}. Using only 25.0 g gives −27.2; it is negative because the temperature rose.

QUICK CHECK

$\Delta_c H$ of ethene is −1411 kJ mol^{-1} and $\Delta_c H$ of ethanol is −1367 kJ mol^{-1}. What is ΔH for $C_2H_4(g) + H_2O(l) \rightarrow C_2H_5OH(l)$?

- −44 kJ mol^{-1}
- +44 kJ mol^{-1}
- −2778 kJ mol^{-1}
- −330 kJ mol^{-1}

ΔH = $\Sigma\Delta_c H$(reactants) − $\Sigma\Delta_c H$(products) = (−1411 + 0) − (−1367) = −44 kJ mol^{-1}. Liquid water is already fully burned, so its enthalpy of combustion is zero. Subtracting the wrong way round gives +44.

QUICK CHECK

Using mean bond enthalpies H–H = 436, Cl–Cl = 243 and H–Cl = 432 kJ mol^{-1}, what is ΔH for $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$?

- +247 kJ mol^{-1}
- +185 kJ mol^{-1}
- −185 kJ mol^{-1}
- −1111 kJ mol^{-1}

Bonds broken = 436 + 243 = 679; bonds made = 2 × 432 = 864. ΔH = 679 − 864 = −185 kJ mol^{-1}. Forgetting that two moles of H–Cl form gives +247.

QUICK CHECK

Why does a calculation with mean bond enthalpies usually give a slightly different ΔH from a Hess cycle using enthalpies of formation?

- Bond breaking is exothermic, so the signs are reversed.
- Heat is lost to the surroundings during the calculation.
- Hess's law does not apply to gases.
- Mean values are averages over many compounds, not the actual values in these molecules.

Each mean bond enthalpy is averaged over many different molecules, so it is only approximately right for any particular molecule. (Bond breaking is endothermic, and heat loss affects experiments, not calculations.)

QUICK CHECK

Which of these enthalpy changes is always positive?

- Standard enthalpy change of combustion
- Standard enthalpy change of atomisation
- Standard enthalpy change of neutralisation
- Standard enthalpy change of formation

Making gaseous atoms from an element always means breaking bonds, which needs energy. Combustion and neutralisation are always exothermic. Formation is usually exothermic, but some compounds, such as ethene (+52 kJ mol^{-1}), have positive values.

QUICK CHECK

A student measures $\Delta_c H$ of propan-1-ol with a spirit burner and a copper can and gets −1150 kJ mol^{-1}. The data book value is −2021 kJ mol^{-1}. Which is the main reason for the difference?

- The specific heat capacity of water is too large.
- The water in the can was not stirred.
- Much of the heat from the flame escaped to the air instead of heating the water.
- The propan-1-ol contained too much oxygen.

Heat loss from the flame is by far the largest error in combustion calorimetry, with incomplete combustion next. Both make the measured value much less exothermic than the true value.

AS · Unit 2 · WCH12 Energetics, Group Chemistry, Halogenoalkanes and Alcohols

Intermolecular forces

Covalent bonds hold the atoms together inside a molecule. Much weaker forces act between neighbouring molecules, and those forces decide whether a substance is a gas, a liquid or a solid, how hot it must be to boil, and what it will dissolve in. This topic explains the three kinds of intermolecular force, how to tell which ones a substance has, and how to use them to explain real data.

Forces between molecules

When a simple molecular substance melts or boils, its molecules move apart, but the molecules themselves stay intact. Boiling water gives steam made of H_2O molecules, not hydrogen and oxygen. So the energy needed to boil a molecular substance is the energy needed to overcome the **intermolecular forces**, and the stronger those forces are, the higher the boiling temperature.

Force	Between which molecules?	Caused by	Relative strength
London forces (instantaneous dipole–induced dipole)	all molecules, and atoms of the noble gases	temporary, constantly changing dipoles	weak for small molecules; grow with the number of electrons
Permanent dipole–dipole forces	polar molecules	permanent δ+ and δ− charges lining up	usually smaller than the London forces in the same molecule
Hydrogen bonds	molecules with H bonded to N, O or F	a strongly δ+ hydrogen attracted to a lone pair on N, O or F	the strongest intermolecular force

KEY IDEA

All three are far weaker than covalent bonds. Even a hydrogen bond is only a small fraction of the strength of the O–H bond inside a water molecule. Forces add up: a molecule that forms hydrogen bonds also has London forces and permanent dipole–dipole forces.

London forces

The electrons in an atom or molecule are always moving. On average they are spread evenly, but at any instant there may be slightly more electron density on one side than the other. That side is momentarily δ−, the other side

δ+: an **instantaneous dipole**.

This dipole affects the next molecule. Its δ− side repels the electrons of the neighbour, and its δ+ side attracts them, so the neighbour's electron cloud shifts too: an **induced dipole**. The δ+ of one molecule now faces the δ− of the next, and they attract. A moment later the dipoles have changed, but the induced dipoles always follow the instantaneous one, so on average there is a net attraction. These attractions are **London forces**. They act between **all** molecules, including non-polar ones and single atoms of the noble gases.

Two things control the strength of London forces:

- **The number of electrons.** More electrons means a larger electron cloud that is held less tightly and distorts more easily, so the instantaneous and induced dipoles are larger. This is why boiling temperatures rise down group 0 and down group 7, and along any homologous series.
- **The area of contact between molecules.** Long, thin molecules can lie alongside each other and touch along their whole length. Compact, branched or nearly spherical molecules touch over a smaller area, so for the same number of electrons the London forces are weaker. (See the alkane isomers on the alkanes page.)

Halogen	Electrons per molecule	Boiling temperature	State at room temperature
F_2	18	−188 °C	gas
Cl_2	34	−34 °C	gas
Br_2	70	59 °C	liquid
I_2	106	184 °C	solid

Permanent dipole–dipole forces

When two atoms with different electronegativities share electrons, the bonding pair is pulled towards the more electronegative atom. The bond is **polar**, with a permanent δ− on one end and δ+ on the other. A molecule is **polar** if it has polar bonds whose dipoles do not cancel out.

- HCl, H_2S, propanone and halogenoalkanes such as chloromethane are polar molecules.
- CO_2 and CCl_4 have polar bonds, but their shapes are symmetrical (linear and tetrahedral), so the bond dipoles cancel. These molecules are **non-polar** and have London forces only.

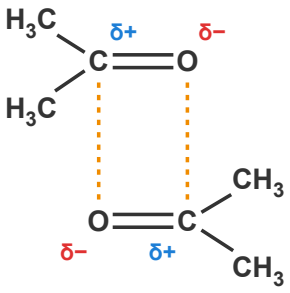
Polar molecules line up so that the δ+ end of one is near the δ− end of the next. The attraction between these permanent dipoles is a **permanent dipole–dipole force**. It acts in addition to the London forces.

hydrogen chloride

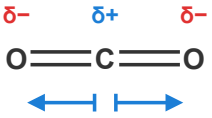


δ− of one molecule attracts δ+ of the next

propanone



carbon dioxide



bond dipoles cancel:
a non-polar molecule

Orange dotted lines: permanent dipole–dipole attractions between polar molecules. CO_2 has polar C=O bonds, but its linear shape makes the two dipoles cancel, so it has London forces only.

COMMON MISTAKE

Do not assume the dipole–dipole forces are the biggest part. For most polar molecules the London forces are larger. HCl, HBr and HI become **less** polar down the group, yet their boiling temperatures **rise**, because the number of electrons, and so the strength of the London forces, increases.

Hydrogen bonds

A **hydrogen bond** is a strong intermolecular attraction between a hydrogen atom bonded to nitrogen, oxygen or fluorine and a **lone pair** on a nitrogen, oxygen or fluorine atom in another molecule. Two features make it much stronger than an ordinary dipole–dipole force:

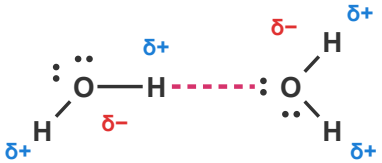
- N, O and F are the most electronegative elements, so the N–H, O–H or F–H bond is very polar and the hydrogen carries a large δ+ charge.
- A hydrogen atom has no inner electrons: pulling its one electron away leaves an almost bare proton, which is tiny, so the lone pair of another molecule can get very close to it.

hydrogen fluoride



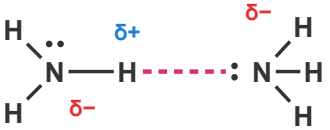
F–H···F: 180°

water



O–H···O: 180°

ammonia



N–H···N: 180°

hydrogen bond, from δ+ H to a lone pair

In each case the hydrogen bond (pink) runs from a δ+ hydrogen to a lone pair on N, O or F, in a straight line. Only one hydrogen bond is drawn in each pair; real liquids form a network of them.

EXAM TIP

Drawing hydrogen bonds for full marks

Show the δ+ on the H and the δ− on the N, O or F atoms; draw the **lone pair** on the atom accepting the hydrogen bond; draw the hydrogen bond as a dashed line from the H to that lone pair; and make the X–H···X arrangement a straight line, **180°**. It is linear because there are two pairs of electrons around the hydrogen (the bonding pair and the lone pair it is attracted to), and they repel each other to be as far apart as possible.

To **predict** whether a substance forms hydrogen bonds, look for an H atom bonded directly to N, O or F. Water, ammonia, hydrogen fluoride, alcohols (–OH), carboxylic acids (–COOH) and amines ($–NH_2$) all qualify. Fluoromethane, CH_3F, does not: its hydrogens are bonded to carbon. Neither do propanone, ethers or aldehydes, although their oxygen lone pairs can **accept** hydrogen bonds from water, which helps them dissolve.

The anomalous properties of water

Unusually high boiling temperature

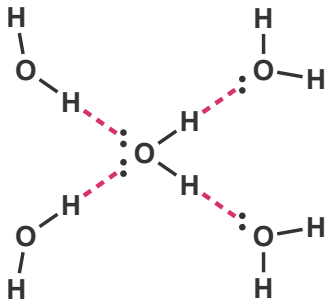
Down each group of the periodic table the hydrides get more electrons, so London forces increase and boiling temperatures rise. Water, ammonia and hydrogen fluoride break this pattern: they have the fewest electrons in their groups, yet boil far higher than the next hydrides down, because they form hydrogen bonds. Methane, in group 14, has no hydrogen bonds and fits the trend.

Why does water boil higher than both HF and NH_3? Count the hydrogen bonds each molecule can make. A water molecule has **two** δ+ hydrogens and **two** lone pairs on oxygen, so on average it forms two hydrogen bonds (and takes part in up to four). HF has three lone pairs but only one hydrogen, and NH_3 has three hydrogens but only one lone pair, so each forms on average only one hydrogen bond per molecule. The hydrogen bonds in HF are individually stronger than those in NH_3, because F is more electronegative than N, so HF boils higher than NH_3.

Ice is less dense than water

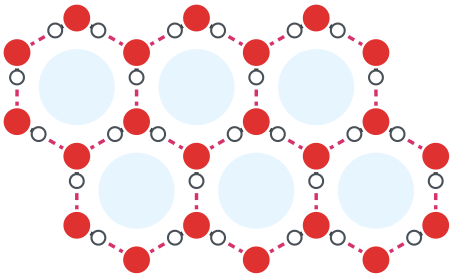
Nearly every substance is denser as a solid than as a liquid. Water is the exception, which is why ice floats.

one molecule, four hydrogen bonds



in three dimensions the four bonds point to the corners of a tetrahedron

oxygen atoms in ice: an open lattice



red: O · white: H · pink dashes: hydrogen bonds
the pale circles mark the empty space in each ring

Left: in ice each molecule donates two hydrogen bonds and accepts two. Right: the hydrogen bonds lock the oxygen atoms into hexagonal rings with a lot of empty space (each oxygen also bonds to a molecule above or below the page). Melting breaks some hydrogen bonds and the molecules move closer together.

Why ice floats

1. In ice, every water molecule is hydrogen-bonded to four others, arranged tetrahedrally around it: two bonds through its own hydrogens and two through the lone pairs on its oxygen.
2. The hydrogen bonds have a fixed length and direction, so they hold the molecules in a rigid, **open lattice** with a lot of empty space, and the molecules further apart than in liquid water.
3. When ice melts, some hydrogen bonds break, the open lattice collapses, and the molecules can pack closer together. So the same mass takes up less volume: liquid water is denser than ice.

Ice floating has a big consequence: lakes freeze from the top down, and the ice layer insulates the water below, so aquatic life survives the winter.

Explaining trends in boiling temperatures

Whenever you explain a boiling temperature, decide which forces are present, decide which one makes the difference, and say that more energy is needed to overcome stronger forces **between the molecules**.

Alkanes: chain length and branching

Alkanes are non-polar, so they have London forces only. Each extra CH_2 adds 8 electrons, so boiling temperatures rise steadily with chain length. Branched isomers have the same number of electrons but a smaller area of contact,

so they boil lower: pentane boils at 36 °C, 2,2-dimethylpropane at 10 °C. The alkanes page has an interactive plot of both effects.

Alcohols are much less volatile than alkanes

A **fair** comparison uses molecules with the same number of electrons, so that the London forces are about the same. Then any difference comes from the other forces. Ethanol and propane both have 26 electrons, but ethanol boils at 78 °C and propane at −42 °C. Ethanol molecules form hydrogen bonds through their –OH groups, and a lot more energy is needed to overcome them. "Low volatility" just means the liquid does not evaporate easily, which goes with a high boiling temperature.

The hydrogen halides

	HF	HCl	HBr	HI
Electrons per molecule	10	18	36	54
Boiling temperature	20 °C (293 K)	−85 °C (188 K)	−67 °C (206 K)	−35 °C (238 K)

Explaining the boiling temperatures of HF to HI

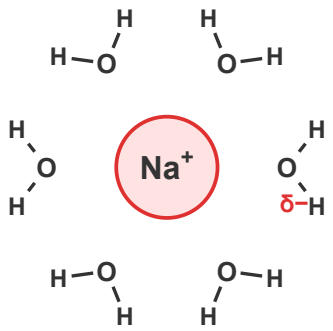
1. **HCl → HBr → HI:** the number of electrons increases (18, 36, 54), so the London forces between the molecules get stronger.
2. The molecules actually become **less** polar down the group, because the difference in electronegativity between H and the halogen gets smaller. So the permanent dipole–dipole forces get weaker.
3. The boiling temperature still rises, so the increase in London forces must outweigh the decrease in dipole–dipole forces. More energy is needed to separate the molecules.
4. **HF** has the fewest electrons, yet it has the highest boiling temperature, because HF molecules form **hydrogen bonds** (H bonded to F, with lone pairs on F). Hydrogen bonds are much stronger than the other intermolecular forces, so far more energy is needed to overcome them.

Solvents and solubility

For a substance to dissolve, its particles must be separated from each other, and the solvent molecules must be pushed apart to make room. Both steps need energy. That energy is supplied by forming new attractions between the solute particles and the solvent molecules. A substance dissolves well when the new solute–solvent attractions are similar in strength to the ones that were broken.

Ionic compounds in water

Water is very polar. When an ionic solid dissolves, water molecules cluster round each ion: the δ− oxygen atoms point towards positive ions, and the δ+ hydrogen atoms point towards negative ions. The ions are said to be **hydrated**. The energy released by these ion–dipole attractions helps to pay for breaking up the ionic lattice. Small ions and ions with larger charges (higher charge density) attract water molecules more strongly and are more strongly hydrated.



δ− oxygen atoms face the positive ion

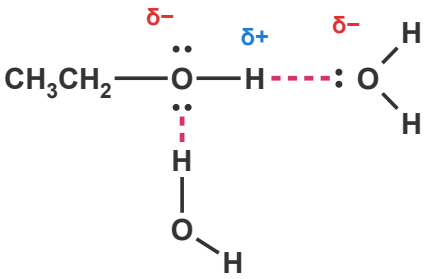
δ+ hydrogen atoms face the negative ion

Hydrated ions. The water molecules turn so that their oppositely charged end faces the ion. The attraction between an ion and these polar molecules releases the energy that breaks up the lattice.

Alcohols in water

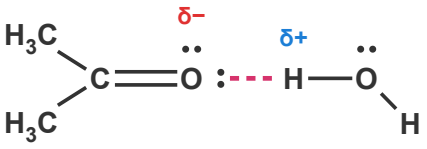
Methanol, ethanol and propanol mix with water in any proportions, because their –OH groups form hydrogen bonds with water molecules, replacing the hydrogen bonds that were broken. As the hydrocarbon chain gets longer, solubility falls. The non-polar chain cannot hydrogen bond, and it gets in the way of the hydrogen bonds between water molecules, so hexan-1-ol is only slightly soluble.

ethanol and water



ethanol both gives and accepts hydrogen bonds

propanone and water



propanone can only accept a hydrogen bond

Hydrogen bonds (pink) between water and organic molecules. Ethanol can form them both ways, through its O–H hydrogen and through its oxygen lone pairs. Propanone has no O–H, but water can still hydrogen bond to its oxygen, which is why propanone mixes with water.

Compounds that do not dissolve in water

A compound that cannot form hydrogen bonds with water is usually insoluble, even if it is polar. Halogenoalkanes such as 1-chlorobutane are polar, but the dipole–dipole and London attractions they could form with water are much weaker than the hydrogen bonds between water molecules that would have to break. Water molecules stay hydrogen-bonded to each other, and the halogenoalkane forms a separate layer. Non-polar substances such as hexane are insoluble in water for the same reason.

Non-aqueous solvents

Substances tend to dissolve in solvents whose intermolecular forces are similar to their own ("like dissolves like"). Iodine, which has only London forces, dissolves readily in hexane, which also has only London forces, but hardly at all in water. Propanone is a very useful solvent because it has both characters: its CH_3 groups form London forces with non-polar substances, and its polar C=O group attracts polar molecules and accepts hydrogen bonds from water, so it mixes with water too.

EXAM TIP

In a solubility answer, name the forces broken in the solute, the forces broken in the solvent, and the new forces formed between them, then compare their strengths. "Hydrogen bonds form between the –OH group of ethanol and water molecules" earns more than "ethanol is polar".

Quick checks

QUICK CHECK

What happens when liquid bromine, Br_2, boils?

- Br–Br covalent bonds break, forming bromine atoms.
- London forces between Br_2 molecules are overcome.
- Hydrogen bonds between Br_2 molecules are broken.
- Permanent dipole–dipole forces between Br_2 molecules are overcome.

Boiling separates whole molecules, so no covalent bonds break. Br_2 is non-polar, so the only intermolecular forces are London forces.

QUICK CHECK

Why do the boiling temperatures of the noble gases increase from helium to xenon?

- The atoms become more polar.
- The covalent bonds between the atoms get stronger.
- The atoms have more electrons, so the instantaneous and induced dipoles are larger and the London forces stronger.
- Heavier atoms move more slowly, so they cannot escape from the liquid.

Noble gas atoms are single, non-polar atoms with no covalent bonds between them. Their only attraction is London forces, which increase with the number of electrons.

QUICK CHECK

Which of these substances forms hydrogen bonds between its own molecules?

- fluoromethane, CH_3F
- propanone, CH_3COCH_3
- methylamine, CH_3NH_2
- hydrogen sulfide, H_2S

Methylamine has H atoms bonded to nitrogen, which also carries a lone pair. In CH_3F and propanone no H is bonded to N, O or F; sulfur is not electronegative enough for H_2S to hydrogen bond.

QUICK CHECK

Which statement about hydrogen bonding in ice is correct?

- Each water molecule is hydrogen-bonded to two others.
- The hydrogen bonds hold the molecules further apart than in liquid water, so ice is less dense.
- The O–H···O angle at each hydrogen bond is 104.5°.
- When ice melts, the covalent O–H bonds break.

In ice each molecule is hydrogen-bonded to four others in an open tetrahedral lattice. The O–H···O arrangement is linear (180°); 104.5° is the H–O–H bond angle inside one molecule. Melting only breaks some hydrogen bonds.

QUICK CHECK

HCl boils at 188 K and HBr at 206 K. Which explanation is correct?

- HBr is more polar, so its dipole–dipole forces are stronger.
- HBr forms hydrogen bonds but HCl does not.
- The H–Br bond is stronger than the H–Cl bond.
- HBr has more electrons, so its London forces are stronger, outweighing its smaller dipole.

Bromine is less electronegative than chlorine, so HBr is actually less polar. Neither forms hydrogen bonds, and bond strength inside the molecule is irrelevant to boiling. The extra electrons strengthen the London forces enough to raise the boiling temperature.

QUICK CHECK

Butane (34 electrons) boils at −1 °C and propan-1-ol (34 electrons) at 97 °C. What is the main reason?

- Propan-1-ol molecules form hydrogen bonds; butane molecules only have London forces.
- Propan-1-ol has more electrons, so stronger London forces.
- Butane is branched, so its molecules have less contact.
- Propan-1-ol has stronger covalent bonds.

The two have the same number of electrons, so their London forces are similar. The difference is the hydrogen bonding through the –OH group, which needs much more energy to overcome. Butane is a straight chain, not branched.

QUICK CHECK

Why is 1-chlorobutane almost insoluble in water, even though its molecules are polar?

- It reacts with water.
- Its molecules are too large to fit between water molecules.
- It cannot form hydrogen bonds with water, and its weaker attractions to water do not make up for breaking hydrogen bonds between water molecules.
- Its C–Cl bond is non-polar.

To dissolve, it would have to break into water's hydrogen-bonded network. The dipole–dipole and London attractions it can form with water are too weak to compensate, so it stays as a separate layer.

QUICK CHECK

In a solution of sodium chloride, how are the water molecules arranged around the ions?

- H atoms point towards Na^+; O atoms point towards Cl^-.
- O atoms point towards Na^+; H atoms point towards Cl^-.
- O atoms point towards both ions.
- Water molecules form covalent bonds to both ions.

The oxygen of water is δ−, so it is attracted to the positive sodium ion; the hydrogens are δ+, so they are attracted to the negative chloride ion. These ion–dipole attractions are what hydration means.

AS · Unit 2 · WCH12 Energetics, Group Chemistry, Halogenoalkanes and Alcohols

Redox chemistry and groups 1, 2 and 7

Oxidation numbers turn electron transfer into simple bookkeeping, and half-equations let you build any redox equation. With those tools you can explain the chemistry of the s-block metals and the halogens: why group 2 metals get more reactive down the group while the halogens get less reactive, why carbonates become harder to decompose, and how a few test-tube reactions identify an unknown ion. The topic finishes with acid–base titrations and measurement uncertainty.

Oxidation numbers

In an ionic compound it is obvious which atom has gained electrons and which has lost them. In a covalent molecule the electrons are shared, so it is less clear. An **oxidation number** is a way of keeping track anyway: it is the charge an atom would have if all of its bonds were fully ionic, with the shared electrons given to the more electronegative atom. Oxidation numbers are written with the sign first, +2 or −1.

Rules for working them out

- **Uncombined elements** have an oxidation number of 0: Zn, O_2, Cl_2, S_8.
- **Simple ions** have an oxidation number equal to their charge: Mg^{2+} is +2, Cl^- is −1.
- In a **compound** the oxidation numbers add up to 0. In a **polyatomic ion** they add up to the charge on the ion.
- **Elements that nearly always have the same value:** group 1 metals +1; group 2 metals +2; aluminium +3; fluorine −1; hydrogen +1 (but −1 in metal hydrides such as NaH); oxygen −2 (but −1 in peroxides such as H_2O_2, and +2 in OF_2); chlorine, bromine and iodine −1 (except when combined with oxygen or fluorine).

Elements such as carbon, nitrogen, sulfur, the transition metals, and the halogens in their oxygen compounds can take many oxidation numbers. You work theirs out from the fixed ones.

Working out oxidation numbers

1. **Iron in $FeCl_3$.** Each chlorine is −1, so three chlorines give −3. The compound is neutral, so iron must be **+3**.

   The oxidation number is for one atom of the element. Chlorine in $CaCl_2$ is −1, not −2.

2. **Sulfur in SO_4^{2-}.** Let sulfur be x. Four oxygens at −2 each, and the total must equal the charge on the ion:

$$x + 4(-2) = -2 \quad \Rightarrow \quad x = +6$$

3 **Chromium in $K_2Cr_2O_7$.** Potassium is +1 and there are two chromium atoms:

$$2(+1) + 2x + 7(-2) = 0 \quad \Rightarrow \quad 2x = +12 \quad \Rightarrow \quad x = +6$$

4 **Nitrogen in NH_4^+.** $x + 4(+1) = +1$, so nitrogen is **−3**.

Naming compounds with oxidation numbers

When an element can have more than one oxidation number, the name shows which one with a Roman numeral. The IUPAC names of the oxyanions (ions containing oxygen) all end in "-ate", with the oxidation number of the central atom in brackets.

Formula	Name	Formula	Name
$FeCl_2$	iron(II) chloride	$FeCl_3$	iron(III) chloride
MnO_2	manganese(IV) oxide	$KMnO_4$	potassium manganate(VII)
K_2SO_4	potassium sulfate(VI)	K_2SO_3	potassium sulfate(IV)
$NaNO_3$	sodium nitrate(V)	$NaNO_2$	sodium nitrate(III)
$NaClO$	sodium chlorate(I)	$NaClO_3$	sodium chlorate(V)

Oxidation and reduction

KEY IDEA

Oxidation is loss of electrons, which **increases** the oxidation number. **Reduction** is gain of electrons, which **decreases** the oxidation number. Electrons lost by one species must be gained by another, so the two always happen together: a **redox** reaction.

- **Oxidising agent (oxidant):** an electron **acceptor**. It takes electrons from another species, so it oxidises that species and is itself **reduced**.
- **Reducing agent (reductant):** an electron **donor**. It gives electrons to another species and is itself **oxidised**.

Metals have few outer electrons and lose them to form positive ions, so in their reactions metals are oxidised and act as reducing agents: $Zn \rightarrow Zn^{2+} + 2e^-$. Non-metals tend to gain electrons to form negative ions, so they are reduced and act as oxidising agents: $Cl_2 + 2e^- \rightarrow 2Cl^-$.

EXAM TIP

When you are asked to identify an oxidising or reducing agent, give the full name or formula of the species, such as "chlorine, Cl_2" or "iodide ions", not just "Cl" or "the iodine". When you say what has been oxidised, give the element and its change: "lithium is oxidised from 0 to +1".

Track the oxidation numbers of every element that changes:

Reaction	Oxidised	Reduced
$4Li(s) + O_2(g) \rightarrow 2Li_2O(s)$	Li: 0 → +1	O: 0 → −2
$WO_3(s) + 3H_2(g) \rightarrow W(s) + 3H_2O(g)$	H: 0 → +1	W: +6 → 0
$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$	Mg: 0 → +2	H: +1 → 0
$2Sr(NO_3)_2(s) \rightarrow 2SrO(s) + 4NO_2(g) + O_2(g)$	O: −2 → 0 (only some of the O atoms)	N: +5 → +4

Metals with dilute acids are redox reactions: metal + acid → salt + hydrogen. The metal is oxidised and hydrogen ions are reduced, so the ionic equation is simply $Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$. With sulfuric acid, $Fe(s) + H_2SO_4(aq) \rightarrow FeSO_4(aq) + H_2(g)$. You would see effervescence (fizzing, as hydrogen is given off) and the metal dissolving.

Disproportionation

DEFINITION

Disproportionation

A redox reaction in which the same element in a single species is simultaneously oxidised and reduced.

Chlorine reacting with water is the classic example. Half of the chlorine atoms go up from 0 to +1 (in chloric(I) acid, HClO) and half go down from 0 to −1 (in HCl):

$$Cl_2(aq) + H_2O(l) \rightleftharpoons HCl(aq) + HClO(aq)$$

Copper(I) ions do the same in aqueous solution, for example when copper(I) oxide reacts with dilute sulfuric acid: $2Cu^+(aq) \rightarrow Cu(s) + Cu^{2+}(aq)$. One copper goes from +1 to 0, the other from +1 to +2.

Half-equations and full redox equations

A **half-equation** shows just one half of a redox reaction, with the electrons written in. In a reduction half-equation the electrons are on the left (they are gained); in an oxidation half-equation they are on the right. For the reaction of bromine with iodide ions:

$$\text{reduction: } Br_2(aq) + 2e^- \rightarrow 2Br^-(aq)$$

$$\text{oxidation: } 2I^-(aq) \rightarrow I_2(aq) + 2e^-$$

$$\text{overall: } Br_2(aq) + 2I^-(aq) \rightarrow 2Br^-(aq) + I_2(aq)$$

Bromine is the oxidising agent (it accepts electrons); iodide ions are the reducing agent.

Balancing half-equations in acidic solution

When the species changes how much oxygen it contains, as in $MnO_4^- \rightarrow Mn^{2+}$, you balance the oxygen with water and the hydrogen with H^+ ions. Work through these steps in order:

Half-equation for $MnO_4^- \rightarrow Mn^{2+}$

1. Balance the atoms of the element that changes (here one Mn on each side is already balanced), and find its change in oxidation number: Mn goes from +7 to +2, a gain of 5 electrons.

$$\mathrm{MnO_4^- + 5e^- \rightarrow Mn^{2+}}$$

2. Balance oxygen by adding water to the side that needs it. Four O atoms on the left need four H_2O on the right:

$$\mathrm{MnO_4^- + 5e^- \rightarrow Mn^{2+} + 4H_2O}$$

3. Balance hydrogen by adding H^+ ions. Eight H atoms in the water need $8H^+$ on the left:

$$\mathrm{MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O}$$

4. Check that the charges balance. Left: (−1) + (+8) + (−5) = +2. Right: +2. Balanced.

If you only balance the atoms and not the charges, the number of electrons is probably wrong.

Half-equation for $SO_4^{2-} \rightarrow H_2S$

1. Sulfur goes from +6 to −2, a gain of 8 electrons:

$$\mathrm{SO_4^{2-} + 8e^- \rightarrow H_2S}$$

2. Four O atoms on the left, so add $4H_2O$ on the right:

$$\mathrm{SO_4^{2-} + 8e^- \rightarrow H_2S + 4H_2O}$$

3. There are now 2 + 8 = 10 H atoms on the right, so add $10H^+$ on the left:

$$\mathrm{SO_4^{2-} + 10H^+ + 8e^- \rightarrow H_2S + 4H_2O}$$

4. Check the charges. Left: (−2) + (+10) + (−8) = 0. Right: 0. Balanced.

Combining half-equations

Electrons can't appear in an overall equation: every electron released by the reducing agent must be taken up by the oxidising agent. So multiply one or both half-equations until the numbers of electrons are equal, then add them and cancel the electrons (and anything else that appears on both sides).

Manganate(VII) ions oxidising iron(II) ions

1. The two half-equations:

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$$

$$Fe^{2+} \rightarrow Fe^{3+} + e^-$$

2. The reduction takes 5 electrons and the oxidation gives 1, so multiply the iron half-equation by 5:

$$5Fe^{2+} \rightarrow 5Fe^{3+} + 5e^-$$

3. Add the two and cancel the $5e^-$:

$$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$$

Check: charge on the left = −1 + 8 + 10 = +17; on the right = +2 + 15 = +17.

Iodide ions reducing sulfuric acid to hydrogen sulfide

1. Oxidation: $2I^- \rightarrow I_2 + 2e^-$. Reduction: $SO_4^{2-} + 10H^+ + 8e^- \rightarrow H_2S + 4H_2O$.

2. Multiply the iodide half-equation by 4 to release 8 electrons, then add:

$$8I^- + SO_4^{2-} + 10H^+ \rightarrow 4I_2 + H_2S + 4H_2O$$

Charges: left = −8 − 2 + 10 = 0, right = 0.

Groups 1 and 2: trends in properties

Groups 1 and 2 make up the s-block. Their atoms have one (group 1) or two (group 2) electrons in an outer s sub-shell, and nearly all of their chemistry comes from losing those electrons to form M^+ or M^{2+} ions.

- **Atomic radius increases down the group.** Each element has one more occupied shell of electrons than the one above it.
- **First ionisation energy decreases down the group.** The nuclear charge does increase, but the outer electron is in a shell further from the nucleus and is shielded by more inner shells. The larger distance and extra shielding outweigh the larger nuclear charge, so the outer electron is held less strongly and is easier to remove.
- **Reactivity increases down the group.** In their reactions the metals lose their outer electrons, and the total energy needed to remove them (the first and second ionisation energies, for group 2) falls down the group.

Reactions of the group 2 elements

With oxygen

All of the group 2 metals burn in oxygen to form white ionic oxides with high melting points, for example $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$, which happens with a brilliant white flame. Magnesium also reacts slowly with oxygen in the air at room temperature, so magnesium ribbon is covered with a thin layer of MgO. Clean it with emery paper before using it in experiments: the oxide reacts with acids too ($MgO + 2HCl \rightarrow MgCl_2 + H_2O$), but without giving off hydrogen, and this would spoil rate measurements.

With chlorine

The metals react with chlorine when heated to form white ionic chlorides: $Mg(s) + Cl_2(g) \rightarrow MgCl_2(s)$, and in the same way Ca, Sr and Ba give $CaCl_2$, $SrCl_2$ and $BaCl_2$. In each case the metal is oxidised from 0 to +2 and the chlorine is reduced from 0 to −1.

With water

Magnesium reacts only very slowly with cold water, forming a few bubbles of hydrogen over several days:

$$Mg(s) + 2H_2O(l) \rightarrow Mg(OH)_2(s) + H_2(g)$$

Heated magnesium burns in **steam** with a bright white flame, and because the temperature is so high the product is the oxide, not the hydroxide:

$$Mg(s) + H_2O(g) \rightarrow MgO(s) + H_2(g)$$

Calcium, strontium and barium react with **cold water**, more vigorously down the group, to form hydroxides and hydrogen:

$$Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(aq) + H_2(g)$$
$$Ba(s) + 2H_2O(l) \rightarrow Ba(OH)_2(aq) + H_2(g)$$

You would see fizzing, the metal dissolving and the water warming up, all faster down the group. With calcium the solution turns cloudy because calcium hydroxide is only slightly soluble; the further down the group, the less of a precipitate forms. The solutions are alkaline. The group 1 metals react with cold water in a similar way, but more vigorously than the group 2 metals in the same period, again more so down the group: $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$.

Oxides and hydroxides

Oxides with water

The oxides of groups 1 and 2 are ionic and contain the oxide ion, O^{2-}. The oxide ion is a strong base: it accepts a proton from water to form hydroxide ions, $O^{2-} + H_2O \rightarrow 2OH^-$. So these oxides dissolve, or react, to give alkaline solutions.

Oxide	Equation	Approximate pH
sodium oxide	$Na_2O(s) + H_2O(l) \rightarrow 2NaOH(aq)$	14 (all group 1 oxides give strongly alkaline solutions)
calcium oxide	$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq)$	about 12
magnesium oxide	$MgO(s) + H_2O(l) \rightarrow Mg(OH)_2(s)$	about 10

Magnesium hydroxide is only very slightly soluble, so its suspension contains few free OH^- ions and the pH is lower. Down group 2 the hydroxides become more soluble, so their saturated solutions become more alkaline.

Oxides and hydroxides with acids

The oxides and hydroxides are bases, so they neutralise acids to give a salt and water:

$$MgO(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2O(l)$$
$$K_2O(s) + 2HNO_3(aq) \rightarrow 2KNO_3(aq) + H_2O(l)$$
$$Mg(OH)_2(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + 2H_2O(l)$$
$$Ca(OH)_2(aq) + 2HNO_3(aq) \rightarrow Ca(NO_3)_2(aq) + 2H_2O(l)$$

Solubility of group 2 hydroxides and sulfates

KEY IDEA

Down group 2, the **hydroxides become more soluble** and the **sulfates become less soluble**. $Mg(OH)_2$ is almost insoluble and $Ba(OH)_2$ is quite soluble; $MgSO_4$ is soluble and $BaSO_4$ is almost completely insoluble.

- **Magnesium hydroxide** forms a white precipitate when hydroxide ions are added to a solution of magnesium ions: $Mg^{2+}(aq) + 2OH^{-}(aq) \rightarrow Mg(OH)_2(s)$. A suspension of it ("milk of magnesia") is used as an antacid to neutralise excess stomach acid. Because so little dissolves, it is only weakly alkaline and safe to swallow, and unlike a carbonate it does not produce carbon dioxide in the stomach.
- **Calcium hydroxide** (slaked lime) is used to neutralise acidic soils. Its solution is **limewater**, which tests for carbon dioxide: $Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$. The insoluble calcium carbonate makes it turn cloudy (milky).
- **Barium hydroxide** dissolves readily: $Ba(OH)_2(s) + aq \rightarrow Ba^{2+}(aq) + 2OH^{-}(aq)$, giving a strongly alkaline solution.
- **Barium sulfate** is so insoluble that it is the basis of the test for sulfate ions (below). It is also why barium reacts only briefly with dilute sulfuric acid: a layer of insoluble $BaSO_4$ forms on the surface and stops the acid reaching the metal. Calcium and strontium are affected less, because their sulfates are a little more soluble. With hydrochloric or nitric acid the chlorides and nitrates are soluble, so the reaction continues.

Thermal stability of carbonates and nitrates

DEFINITION

Thermal decomposition

The breaking down of a compound into two or more products by heating.

Group 2 carbonates decompose on heating to the metal oxide and carbon dioxide. Group 2 nitrates decompose to the oxide, brown nitrogen dioxide and oxygen. The white solid nitrate often melts to a colourless liquid before it decomposes and then turns back into a white solid (the oxide).

$$MgCO_3(s) \rightarrow MgO(s) + CO_2(g)$$

$$2Ca(NO_3)_2(s) \rightarrow 2CaO(s) + 4NO_2(g) + O_2(g)$$

KEY IDEA

Thermal stability increases down the group. Magnesium carbonate decomposes at around 540 °C, but barium carbonate needs about 1360 °C. The same pattern holds for the nitrates.

The explanation lies in the **polarising power** of the metal ion. The carbonate ion is large, and its electron cloud is easy to distort. A small, highly charged cation has a high **charge density**, so it pulls the nearby electron cloud of the carbonate ion towards itself. This distortion weakens the C–O bonds in the carbonate ion, making it easier to break off CO_2 and leave O^{2-} behind. Down group 2 the ions get bigger (Mg^{2+} 72 pm to Ba^{2+} 135 pm) with the same 2+ charge, so their charge density falls, they polarise the anion less, and a higher temperature is needed. Nitrate ions are affected in the same way, with the N–O bonds weakened.

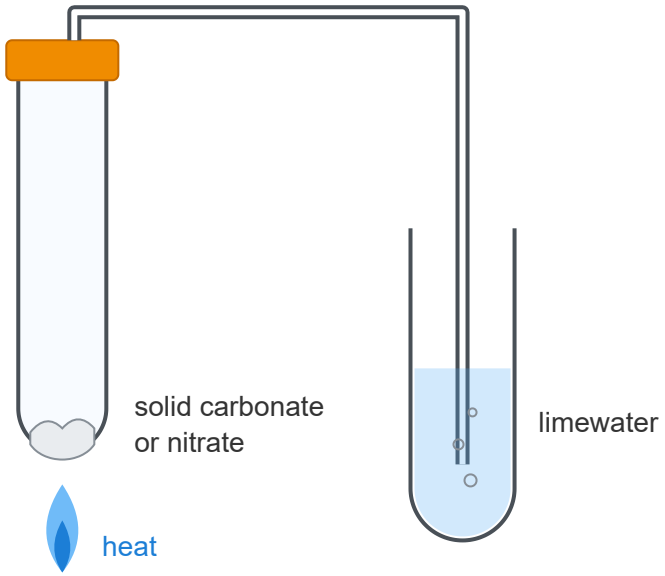
Group 1 compounds are more stable. A 1+ ion has a lower charge density than the 2+ ion next to it, so it hardly polarises the anion. Most group 1 carbonates do not decompose at the temperature of a Bunsen burner, and most

group 1 nitrates decompose only as far as the nitrite (nitrate(III)) and oxygen, with no brown gas:

$$2NaNO_3(s) \rightarrow 2NaNO_2(s) + O_2(g)$$

Lithium is the exception. Li^+ is so small that its charge density is high enough to behave like a group 2 ion, so its compounds decompose in the same way as group 2 compounds:

$$Li_2CO_3(s) \rightarrow Li_2O(s) + CO_2(g)$$
$$4LiNO_3(s) \rightarrow 2Li_2O(s) + 4NO_2(g) + O_2(g)$$



Comparing thermal stability. Heat the same number of moles of each carbonate in the same way and time how long the limewater takes to turn cloudy. For nitrates, time how long it takes for brown NO_2 to appear, or for the gas to relight a glowing splint.

To make the comparison fair, use the same **amount in moles** of each compound, the same volume of limewater, the same Bunsen flame and the same height of the tube above it. Remove the delivery tube from the limewater before you stop heating, or the limewater can be sucked back into the hot tube.

Flame tests

Many group 1 and 2 compounds give a characteristic colour to a flame, which identifies the metal ion.

- Clean a nichrome wire loop by dipping it in concentrated hydrochloric acid and holding it in a roaring (blue) Bunsen flame until it gives no colour. Nichrome is unreactive and does not colour the flame itself.
- Dip the loop in the acid again, then into the powdered solid, and hold it at the edge of the flame. Look at the colour.

Ion	Flame colour	Ion	Flame colour
Li^+	red (crimson)	Mg^{2+}	no colour

Na^+	yellow-orange	Ca^{2+}	brick red (orange-red)
K^+	lilac	Sr^{2+}	red (crimson)
Rb^+	red-violet	Ba^{2+}	pale (apple) green
Cs^+	blue-violet		

Where does the colour come from? The heat of the flame gives some electrons in the metal ions enough energy to jump to **higher energy levels**. These excited electrons are unstable and soon fall back to lower levels, giving out the energy difference as light. Because the energy levels are fixed for each element, the energy gaps, and so the frequencies of light emitted, are characteristic of that element. Magnesium compounds give no colour because the energy they emit corresponds to light outside the visible part of the spectrum.

Testing for ions

- **Ammonium ion, NH_4^+:** warm the substance with sodium hydroxide solution. Ammonia is given off, $NH_4^+(aq) + OH^-(aq) \rightarrow NH_3(g) + H_2O(l)$. It has a pungent smell and turns damp red litmus paper blue.
- **Carbonate, CO_3^{2-}, and hydrogencarbonate, HCO_3^-:** add a dilute acid. Effervescence of a gas that turns limewater cloudy shows carbon dioxide: $2H^+ + CO_3^{2-} \rightarrow H_2O + CO_2$, or $H^+ + HCO_3^- \rightarrow H_2O + CO_2$.
- **Sulfate, SO_4^{2-}:** add dilute hydrochloric acid, then barium chloride solution. A white precipitate of barium sulfate shows sulfate ions: $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$. The acid is added first to remove any carbonate ions, which would otherwise give a white precipitate of barium carbonate and a false positive. Sulfuric acid cannot be used for this, because it contains sulfate ions itself.
- **Halide ions:** acidified silver nitrate solution, described in the group 7 section below.
- **Group 1 and 2 metal ions:** flame tests.

Group 7: the halogens

The halogens exist as diatomic molecules, X_2, held together by a single covalent bond. At room temperature fluorine is a very pale yellow gas, chlorine a pale green gas, bromine a dark red-brown liquid that gives off orange-brown vapour, and iodine a shiny grey-black solid that turns into a purple vapour when warmed. All are toxic.

Physical trends

- **Melting and boiling temperatures increase down the group.** The molecules have more electrons, so the London forces between them are stronger and more energy is needed to separate them. That is why the state changes from gas to liquid to solid down the group. Astatine, below iodine, would be expected to be a black solid.
- **Electronegativity decreases down the group.** The atoms get larger, with more shells of electrons, so a bonding pair is further from the nucleus and more shielded from it, and is attracted less strongly.

Oxidising power

When a halogen reacts, its atoms gain an electron each to become halide ions: $X_2 + 2e^- \rightarrow 2X^-$. So halogens are oxidising agents. Their **oxidising power decreases down the group**: the atoms get bigger, so an incoming electron is further from the nucleus and more shielded from it, and is attracted less strongly. Fluorine is the strongest oxidising agent of all the elements.

The clearest evidence is **displacement**: a halogen oxidises the halide ions of any halogen *below* it in the group. Chlorine displaces bromine and iodine; bromine displaces iodine; iodine displaces neither.

$$Cl_2(aq) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$$

$$Cl_2(aq) + 2I^-(aq) \rightarrow 2Cl^-(aq) + I_2(aq)$$

$$Br_2(aq) + 2I^-(aq) \rightarrow 2Br^-(aq) + I_2(aq)$$

In water the colours can be hard to tell apart, because dilute bromine and dilute iodine both look yellow-brown. Shaking the mixture with a little **cyclohexane** helps: the halogen dissolves much better in this non-polar solvent, which floats on the water as the upper layer, and the colours there are distinct.

Halogen	Colour in water	Colour in cyclohexane
chlorine	very pale green (almost colourless)	very pale green (almost colourless)
bromine	yellow to orange	orange to red-brown
iodine	yellow-brown to brown	violet (purple)

Halogens oxidise metals too, the metal always being oxidised: $2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$ and $2Fe(s) + 3Cl_2(g) \rightarrow 2FeCl_3(s)$. Chlorine and bromine are strong enough oxidising agents to turn iron(II) ions into iron(III) ions, $Cl_2(aq) + 2Fe^{2+}(aq) \rightarrow 2Cl^-(aq) + 2Fe^{3+}(aq)$, but iodine is not. In fact the reaction goes the other way for iodine: iron(III) ions oxidise iodide ions, $2Fe^{3+}(aq) + 2I^-(aq) \rightarrow 2Fe^{2+}(aq) + I_2(aq)$.

Chlorine with water and with alkali

Chlorine dissolves in water and partly disproportionates, giving a mixture of hydrochloric acid and chloric(I) acid:

$$Cl_2(aq) + H_2O(l) \rightleftharpoons HCl(aq) + HClO(aq)$$

Universal indicator added to chlorine water first turns red, because both products are acids, and is then bleached colourless by the chloric(I) acid. Chloric(I) acid kills bacteria, which is why chlorine is added to drinking water and swimming pools. Chlorine is toxic, and it can react with organic matter in water to form small amounts of possibly harmful chlorinated compounds, but at the low concentrations used the benefit of preventing water-borne diseases such as cholera far outweighs the risk.

With **cold, dilute** sodium hydroxide, chlorine disproportionates to chloride and chlorate(I). The mixture of sodium chloride and sodium chlorate(I) is household bleach:

$$Cl_2(aq) + 2NaOH(aq) \rightarrow NaCl(aq) + NaClO(aq) + H_2O(l) \quad (Cl: 0 \rightarrow -1 \text{ and } 0 \rightarrow +1)$$

With **hot** sodium hydroxide, the chlorate(I) disproportionates further, so the chlorine ends up as chloride and chlorate(V):

$$3Cl_2(aq) + 6NaOH(aq) \rightarrow 5NaCl(aq) + NaClO_3(aq) + 3H_2O(l) \quad (Cl: 0 \rightarrow -1 \text{ and } 0 \rightarrow +5)$$

Electron bookkeeping confirms the 5 : 1 ratio: five chlorine atoms each gain one electron (5 e^- in total) while one chlorine atom loses five. As an ionic equation, $3Cl_2 + 6OH^- \rightarrow 5Cl^- + ClO_3^- + 3H_2O$. Bromine and iodine react with alkalis in the same ways, which is why their colours fade when sodium hydroxide is added.

Halide ions as reducing agents

The halide ions can give an electron back, $2X^- \rightarrow X_2 + 2e^-$, so they are reducing agents. Their **reducing power increases down the group**: a larger ion holds its outer electrons further from the nucleus, with more shielding, so it loses them more easily. Iodide is the strongest reducing agent of the four; fluoride and chloride are very weak ones. Concentrated sulfuric acid shows the difference clearly, because how far the halide reduces the sulfur in H_2SO_4 (oxidation number +6) depends on its reducing power.

Solid halide + conc. H_2SO_4	Equations	Observations	Sulfur reduced to
NaF, NaCl	$NaCl(s) + H_2SO_4(l) \rightarrow NaHSO_4(s) + HCl(g)$ (NaF gives HF in the same way)	steamy white fumes of HCl (or HF)	not reduced: an acid–base reaction only
NaBr	$NaBr + H_2SO_4 \rightarrow NaHSO_4 + HBr$ $2HBr + H_2SO_4 \rightarrow Br_2 + SO_2 + 2H_2O$	steamy fumes of HBr; orange-brown fumes of bromine; colourless, acidic SO_2	+4 (SO_2)
NaI	$NaI + H_2SO_4 \rightarrow NaHSO_4 + HI$ $2HI + H_2SO_4 \rightarrow I_2 + SO_2 + 2H_2O$ $6HI + H_2SO_4 \rightarrow 3I_2 + S + 4H_2O$ $8HI + H_2SO_4 \rightarrow 4I_2 + H_2S + 4H_2O$	some steamy fumes of HI; black solid and purple vapour of iodine; yellow solid sulfur; smell of bad eggs from H_2S	+4 (SO_2), 0 (S) and −2 (H_2S)

In every case the first step is an acid–base reaction, with sulfuric acid acting as a proton donor to make the hydrogen halide. Chloride and fluoride ions are not strong enough reducing agents to go any further. Bromide and iodide ions then reduce the sulfuric acid, which now acts as an oxidising agent. The steamy fumes appear because the hydrogen halide gas dissolves in moisture in the air.

EXAM TIP

Exam questions often give you the products and ask for the equation. Build it from half-equations: for example $2Br^- \rightarrow Br_2 + 2e^-$ with $H_2SO_4 + 2H^+ + 2e^- \rightarrow SO_2 + 2H_2O$.

Testing for halide ions with silver nitrate

Add dilute **nitric acid** to the solution, then a few drops of silver nitrate solution. The silver halides of chloride, bromide and iodide are insoluble and appear as precipitates of different colours. The nitric acid removes ions such as carbonate, which would otherwise form a precipitate of silver carbonate and confuse the result. (Hydrochloric acid can't be used, because it would add chloride ions.)

Halide	With acidified $AgNO_3(aq)$	Ionic equation	Then add ammonia solution
fluoride	no precipitate (AgF is soluble)	no reaction	—
chloride	white precipitate	$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$	dissolves in dilute ammonia
bromide	cream precipitate	$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$	dissolves in concentrated ammonia only
iodide	pale yellow precipitate	$Ag^+(aq) + I^-(aq) \rightarrow AgI(s)$	does not dissolve, even in concentrated ammonia

The precipitate colours are similar, so the ammonia step confirms the result. Silver chloride dissolves because the silver ions form a soluble complex ion with ammonia: $AgCl(s) + 2NH_3(aq) \rightarrow [Ag(NH_3)_2]^+(aq) + Cl^-(aq)$. The silver halides become less soluble down the group, so silver bromide needs a higher concentration of ammonia and silver iodide doesn't dissolve at all.

Hydrogen halides

HCl, HBr and HI are colourless gases that are very soluble in water. They react with it to form acidic solutions, for example $HCl(g) + H_2O(l) \rightarrow H_3O^+(aq) + Cl^-(aq)$, which is hydrochloric acid. They also react with ammonia gas to form white smoke, a cloud of tiny crystals of the ammonium halide: $NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$. Holding a glass rod dipped in concentrated ammonia solution near the gas is a test for a hydrogen halide.

Acid–base titrations

A titration finds the concentration of a solution by reacting it with a solution of known concentration, a **standard solution**. Core Practicals 3 and 4 cover making a standard solution and using it in a simple acid–base titration.

CORE PRACTICAL

Making a standard solution

- Weigh the solid accurately in a weighing bottle, tip it into a beaker, and reweigh the bottle to find the exact mass transferred (weighing by difference).
- Dissolve the solid in a small volume of distilled water, stirring with a glass rod.
- Pour the solution through a funnel into a volumetric flask (often 250 cm^3). Rinse the beaker, rod and funnel with distilled water and add all the rinsings to the flask, so that none of the solute is lost.
- Add distilled water until the bottom of the meniscus sits on the graduation line, viewed at eye level; use a dropping pipette for the last few drops.

- Stopper the flask and invert it several times so that the concentration is the same throughout.

CORE PRACTICAL

Carrying out a titration

- Rinse the burette with the acid it will contain, the pipette with the alkali, and the conical flask with distilled water only. Rinsing the flask with distilled water does not change the number of moles that will go into it.
- Use a pipette and filler to put 25.0 cm^3 of the alkali into the conical flask, touching the tip on the surface of the liquid at the end. Don't blow out the small drop left in the tip: the pipette is calibrated to allow for it.
- Add a few drops of indicator and stand the flask on a white tile so the colour change is easy to see.
- Fill the burette, making sure the space below the tap (the jet) is full of liquid and has no air bubble. An air bubble that fills during the titration makes the titre too large.
- Read the burette, then add the acid while swirling the flask, slowing to drop-by-drop near the end point. Washing the sides of the flask with a little distilled water near the end makes sure all of the acid reacts.
- Do a rough titration first, then repeat accurately until you have at least two **concordant** titres, within 0.10 cm^3 of each other. Record readings to the nearest 0.05 cm^3 and average only the concordant titres.
- **Safety:** acids and alkalis are corrosive, or irritants when dilute. Wear eye protection and wash any spills off the skin straight away.

Indicator	Colour in alkali	Colour in acid	End point (acid added to alkali)
phenolphthalein	pink	colourless	pink colour just disappears
methyl orange	yellow	red	first permanent orange

Either indicator works for a strong acid with a strong alkali. Phenolphthalein is used when the acid is weak, such as ethanoic acid titrated with sodium hydroxide.

Worked example: ethanoic acid in vinegar

1. 25.0 cm^3 of vinegar is diluted to 250 cm^3 in a volumetric flask. 25.0 cm^3 of 0.100 mol dm^{-3} NaOH needs 22.40 cm^3 of the diluted vinegar for neutralisation. Find the concentration of ethanoic acid in the original vinegar in g dm^{-3}. $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$.

2. Moles of sodium hydroxide:

$$n(\text{NaOH}) = 0.100 \times \frac{25.0}{1000} = 2.50 \times 10^{-3}\ \text{mol}$$

3. The reacting ratio is 1 : 1, so 2.50 × 10^{-3} mol of ethanoic acid was in 22.40 cm^3 of diluted vinegar:

$$c = \frac{2.50 \times 10^{-3}}{0.02240} = 0.1116\ \text{mol dm}^{-3}$$

4. The vinegar was diluted ten times (25.0 cm^3 made up to 250 cm^3), so the original concentration is 1.116 mol dm^{-3}.

5. Convert to g dm^{-3} using $M_r(CH_3COOH) = 60.0$:

$$1.116 \times 60.0 = 67.0 \text{ g dm}^{-3}$$

Worked example: a back titration on an antacid tablet

1. A 0.500 g antacid tablet containing calcium carbonate is added to 50.0 cm^3 of 0.500 mol dm^{-3} hydrochloric acid (an excess). When it has reacted, the mixture is made up to 250 cm^3. 25.0 cm^3 portions need 15.20 cm^3 of 0.100 mol dm^{-3} NaOH. Find the percentage by mass of $CaCO_3$ in the tablet.

   Calcium carbonate is insoluble, so it can't be titrated directly. Instead you react it with a known excess of acid and titrate the acid that is left over.

2. Acid left in each 25.0 cm^3 portion (HCl : NaOH is 1 : 1):

$$n(\text{HCl}) = 0.100 \times \frac{15.20}{1000} = 1.52 \times 10^{-3} \text{ mol}$$

   So the whole 250 cm^3 contained 1.52 × 10^{-2} mol of unreacted HCl.

3. Acid added at the start: $0.500 \times 0.0500 = 2.50 \times 10^{-2}$ mol. Acid used up by the tablet:

$$2.50 \times 10^{-2} - 1.52 \times 10^{-2} = 9.8 \times 10^{-3} \text{ mol}$$

4. $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$, so the moles of $CaCO_3$ are half of this, 4.9 × 10^{-3} mol. Using $M_r(CaCO_3) = 100.1$:

$$m = 4.9 \times 10^{-3} \times 100.1 = 0.490 \text{ g} \qquad \frac{0.490}{0.500} \times 100 = 98\%$$

Worked example: identifying a group 1 carbonate

1. 2.07 g of a group 1 carbonate, M_2CO_3, is dissolved and made up to 250 cm^3. 25.0 cm^3 of this solution reacts with 30.00 cm^3 of 0.100 mol dm^{-3} HCl. $M_2CO_3 + 2HCl \rightarrow 2MCl + H_2O + CO_2$. Identify M.

2. $n(\mathrm{HCl}) = 0.100 \times 0.03000 = 3.00 \times 10^{-3}$ mol, so $n(\mathrm{M_2CO_3}) = 1.50 \times 10^{-3}$ mol in 25.0 cm^3, and 1.50 × 10^{-2} mol in the full 250 cm^3.

3. $$M_\mathrm{r} = \frac{2.07}{1.50 \times 10^{-2}} = 138$$

4. Subtract the carbonate (12.0 + 3 × 16.0 = 60.0) and halve: A_r(M) = (138 − 60) ÷ 2 = 39, so M is **potassium** and the compound is K_2CO_3.

Measurement uncertainty

Every piece of apparatus has a limited resolution. A **reading** is a single judgement on a scale, such as the level in a measuring cylinder or a balance display; its uncertainty is usually half the smallest division. A **measurement** is the difference between two readings, such as a titre (final minus initial burette reading), so it collects the uncertainty of both readings.

Apparatus	Typical uncertainty
2 d.p. balance	± 0.005 g per reading
25.0 cm^3 pipette	± 0.06 cm^3
250 cm^3 volumetric flask	± 0.2 cm^3
burette	± 0.05 cm^3 per reading, so ± 0.10 cm^3 for a titre

Exam questions always tell you the uncertainty to use. Compare different pieces of apparatus using the **percentage uncertainty**:

$$\text{percentage uncertainty} = \frac{\text{uncertainty}}{\text{quantity measured}} \times 100$$

For a titre of 22.40 cm^3, the percentage uncertainty is $\frac{0.10}{22.40} \times 100 = 0.45\%$. To estimate the overall uncertainty in a result, add the percentage uncertainties of all the measurements that went into it.

- **Reducing uncertainty:** use apparatus with a finer scale (a pipette or burette instead of a measuring cylinder, a 3 d.p. balance), or measure a larger quantity. A bigger titre has a smaller percentage uncertainty, so you could use a more concentrated solution in the flask, a larger volume in the flask, or a more dilute solution in the burette.
- **Weigh by difference:** weighing the container before and after transferring a solid gives the mass actually added, even if a little is left behind.
- **Comparing with an accepted value:** work out the percentage difference, $\frac{|\text{experimental} - \text{accepted}|}{\text{accepted}} \times 100$. If it is smaller than the total percentage uncertainty from the apparatus, the difference can be explained by the limits of the equipment. If it is larger, there must be other errors in the method, such as heat loss or an incomplete reaction.

Quick checks

QUICK CHECK

What is the oxidation number of chlorine in $NaClO_3$?

- −1
- +3
- +5
- +7

Sodium is +1 and the three oxygens are −6 in total, so chlorine must be +5 for the compound to be neutral: (+1) + x + (−6) = 0. That is why $NaClO_3$ is called sodium chlorate(V). Chlorine is only −1 when it is not combined with oxygen or fluorine.

QUICK CHECK

In the reaction $3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$, what happens to the chlorine?

- It is both oxidised and reduced.
- It is oxidised only.
- It is reduced only.
- It is neither oxidised nor reduced.

Chlorine starts at 0. In NaCl it is −1 (reduced) and in $NaClO_3$ it is +5 (oxidised). One element in one species going both ways is disproportionation.

QUICK CHECK

Which statement explains why the first ionisation energy decreases down group 2?

- The nuclear charge decreases down the group.
- The outer electrons are in s orbitals.
- There are fewer inner electrons to shield the outer electrons.
- The outer electrons are further from the nucleus and more shielded, which outweighs the greater nuclear charge.

Nuclear charge actually increases down the group, but each element has an extra shell, so the outer electrons are further away and shielded by more inner electrons. The attraction to the nucleus is weaker overall.

QUICK CHECK

Which group 2 carbonate has the highest decomposition temperature, and why?

- $MgCO_3$, because Mg^{2+} has the highest charge density.
- $BaCO_3$, because Ba^{2+} has the lowest charge density and polarises the carbonate ion least.
- $BaCO_3$, because barium is the most reactive metal.
- $MgCO_3$, because its lattice is the strongest.

The large Ba^{2+} ion distorts the carbonate ion's electron cloud least, so the C–O bonds are weakened least and more heat is needed to break the ion up. Mg^{2+} has the highest charge density, so $MgCO_3$ decomposes most easily.

QUICK CHECK

Concentrated sulfuric acid is added to solid sodium bromide. Which products form?

- HBr only
- HBr, Br_2, SO_2 and H_2S
- HBr, Br_2 and SO_2
- Br_2 and H_2S only

Bromide is a strong enough reducing agent to reduce sulfur from +6 to +4 (SO_2), but not as far as H_2S (−2). Only iodide, the strongest reducing agent, gets that far. HBr forms first in the acid–base step.

QUICK CHECK

A solution gives a cream precipitate with acidified silver nitrate. The precipitate dissolves in concentrated ammonia but not in dilute ammonia. Which ion is present?

- Cl^-
- Br^-
- I^-
- F^-

Silver bromide is cream and dissolves only in concentrated ammonia. Silver chloride is white and dissolves in dilute ammonia; silver iodide is pale yellow and insoluble in both. Fluoride gives no precipitate.

QUICK CHECK

A burette reading has an uncertainty of ±0.05 cm^3. What is the percentage uncertainty in a titre of 12.50 cm^3?

- 0.4%
- 0.8%
- 0.05%
- 8%

A titre uses two readings, so its uncertainty is ±0.10 cm^3: 0.10 ÷ 12.50 × 100 = 0.8%. Using only one reading's uncertainty gives 0.4%.

AS · Unit 2 · WCH12 Energetics, Group Chemistry, Halogenoalkanes and Alcohols

Introduction to kinetics and equilibria

Kinetics asks how fast a reaction goes; equilibrium asks how far it goes. This topic explains rates using collision theory and the Maxwell–Boltzmann distribution of molecular energies, shows how catalysts work, and then uses Le Chatelier's principle to predict how a reversible reaction responds when you change the conditions, including the compromises made in industry.

Collision theory

For two particles to react they must first **collide**. But in a gas at room temperature each molecule collides billions of times a second, and if every collision led to a reaction, every reaction would be over almost instantly. Most collisions are unsuccessful: the particles simply bounce apart. A collision only leads to a reaction if the particles have enough energy between them to start breaking the bonds that need to break (and, for many reactions, if they hit each other the right way round).

DEFINITION

Activation energy, E_a

The minimum energy that colliding particles must have for a reaction to take place.

On a reaction profile the activation energy is the height of the energy barrier between the reactants and the top of the hump. It is there for exothermic reactions as well as endothermic ones: that is why a fuel such as methane can sit in air without burning until a spark gives some molecules enough energy to get over the barrier. Once the reaction starts, the energy it releases keeps it going.

So the rate of a reaction depends on **how often** particles collide and on **what proportion** of those collisions have at least the activation energy. Every factor that changes the rate works through one or both of these.

Measuring rates of reaction

DEFINITION

Rate of reaction

The change in concentration (or amount) of a reactant or product per unit time. Its usual unit is mol dm^{-3} s^{-1}.

To follow a reaction you measure something that changes as it goes, at regular time intervals:

- **Volume of gas** given off, collected in a gas syringe or an upturned measuring cylinder (for example magnesium or a carbonate reacting with an acid).
- **Loss of mass** as a gas escapes, with the flask on a balance (with a cotton wool plug to stop acid spray escaping but let the gas out).
- **Colour change**, followed with a colorimeter.
- **Formation of a precipitate.** Sodium thiosulfate and hydrochloric acid slowly form a cloudy precipitate of sulfur: $Na_2S_2O_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + SO_2(g) + S(s) + H_2O(l)$. You time how long it takes for a cross drawn under the flask to disappear. The same amount of sulfur has to form each time to hide the cross, so the rate is proportional to 1/time.

If you plot the concentration of a reactant (or the volume of a product) against time, the **rate at any moment is the gradient of the curve**. The curve is steepest at the start, because that is when the reactants are most concentrated, and it gets shallower as they are used up. To find the rate at a particular time, draw a **tangent** to the curve at that time and work out its gradient. The tangent at t = 0 gives the **initial rate**.

Worked example: initial rate from a tangent

1. Magnesium reacts with excess hydrochloric acid and the hydrogen is collected in a gas syringe. A tangent drawn to the volume–time curve at t = 0 passes through the origin and through the point (20 s, 36 cm^3). Find the initial rate.

2. The gradient of the tangent is the initial rate:

$$\text{gradient} = \frac{\text{change in volume}}{\text{change in time}}$$

Choose two points on the tangent that are far apart: this makes the gradient more precise.

3. To express it in mol s^{-1}, use the molar volume (24.0 dm^3 mol^{-1} = 24 000 cm^3 mol^{-1} at room temperature and pressure):

$$\text{rate} = \frac{\text{gradient}}{\text{molar volume}}$$

When you compare curves from different experiments, read two things from them. The **initial gradient** tells you which reaction is faster. The **final height** tells you how much product was made, which depends only on the amount of the limiting reactant, not on how fast it reacted. A faster reaction with the same amount of limiting reactant reaches the same final volume, just sooner.

The Maxwell–Boltzmann distribution

The molecules in a gas do not all have the same energy. Collisions keep transferring energy between them, so at any instant a few are almost stationary, most have a moderate energy, and a few have a great deal. The **Maxwell–Boltzmann distribution** is the graph of the number (or fraction) of molecules against their energy, at a particular temperature.

- It starts at the **origin**: no molecules have zero energy.
- The peak is at the **most probable energy**, E_{mp}, the energy that the largest number of molecules have.
- The **mean energy** is a little to the right of the peak, because the curve is not symmetrical: its long tail pulls the average up.
- The curve **approaches the energy axis but never meets it**, because there is no maximum energy a molecule can have.
- The **area under the curve** represents the total number of molecules, so it stays the same if only the temperature changes.
- Only the molecules in the tail with **energy at least E_a** can react when they collide. That is the shaded area beyond the E_a line.

EXAM TIP

When you sketch Maxwell–Boltzmann curves in an exam: start exactly at the origin, never let the tail touch the axis, and label the axes "number of molecules" (or "fraction of molecules") and "energy". If you draw a curve for a higher temperature, its peak must be lower and further right, and it must cross the original curve once and stay above it at high energies, so that the areas are the same.

Factors that affect the rate

Concentration and pressure

A higher concentration of a solution, or a higher pressure of a gas, means **more particles in the same volume**. They collide more often, so there are more **successful collisions per second** and the rate increases. The proportion of particles with energy above E_a does not change: the Maxwell–Boltzmann curve has the same shape, just taller, because there are more molecules altogether. If you double the concentration of a reactant, the number of particles in each unit of volume doubles, and the frequency of successful collisions involving it roughly doubles too.

Temperature

At a higher temperature the molecules have more energy on average. The Maxwell–Boltzmann curve flattens and moves to the right, and the number of molecules with energy greater than or equal to E_a increases a lot. This is the main reason the rate rises: a much **greater proportion of collisions are successful**. The molecules also move

faster and collide a little more often, but that is a much smaller effect. For many reactions near room temperature, a rise of just 10 K roughly doubles the rate.

COMMON MISTAKE

"The particles collide more often" is not enough to explain the effect of temperature. Full marks need the idea that more of the particles have energy equal to or greater than the activation energy, so a larger proportion of collisions succeed.

Surface area

When a solid reacts with a liquid or gas, only the particles on its surface can be hit. Breaking the solid into smaller pieces, or a powder, exposes far more surface, so collisions happen more often and the rate increases. The energy of the collisions is unchanged.

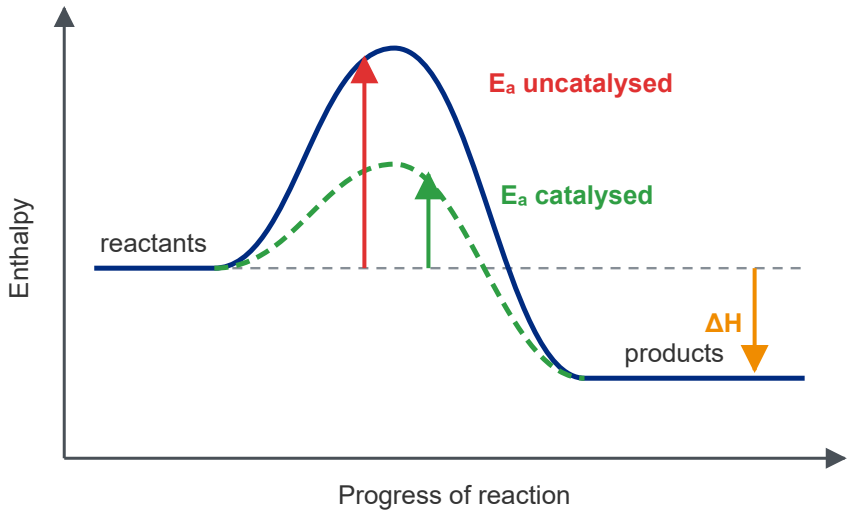
Catalysts

DEFINITION

Catalyst

A substance that increases the rate of a reaction without being used up: it is chemically unchanged at the end. It works by providing an **alternative reaction pathway** with a **lower activation energy**.

With a lower activation energy, a larger proportion of the molecules in the Maxwell–Boltzmann distribution have enough energy to react, so more collisions are successful. Notice what a catalyst does *not* do: it does not give the molecules more energy (the distribution curve is unchanged), and it does not change ΔH, because the reactants and products are the same.



The catalysed route (dashed) has a lower activation energy, but the reactants, the products and ΔH are exactly the same.

Heterogeneous catalysts

A **heterogeneous** catalyst is in a different phase from the reactants. In industry this is usually a solid catalyst with gaseous reactants. The reaction happens on the catalyst's surface: reactant molecules are **adsorbed** (held weakly on the surface), which weakens their bonds and brings them together in a suitable position to react; the product molecules then **desorb** and leave the surface free for more reactants. Because only the surface takes part, the catalyst is used as a powder, pellets, a fine mesh or a thin coating on a honeycomb support, to give the largest possible surface area for the least material. Examples are iron in the Haber process, vanadium(V) oxide in the Contact process, and platinum, palladium and rhodium in catalytic converters.

Why catalysts matter

- **Lower temperatures:** a reaction can run at a useful rate at a lower temperature, so less fuel is burned to heat it. That saves money and cuts carbon dioxide emissions.
- **Better processes:** a catalyst can make a different route possible, for example one with a higher atom economy or fewer steps, which uses fewer raw materials and makes less waste.
- **Reuse:** because the catalyst is not used up, a small amount can be used again and again.

Reversible reactions and dynamic equilibrium

Many reactions are **reversible**: the products can react together to re-form the reactants. The sign ⇌ shows this, for example $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$. Start with just nitrogen and hydrogen in a sealed container: at first only the forward reaction can happen, and it is fast. As ammonia builds up, the backward reaction starts and speeds up, while the forward reaction slows down because the reactants are being used up. Eventually the two rates become equal.

DEFINITION

Dynamic equilibrium

The state of a reversible reaction in a **closed system** when the rate of the forward reaction equals the rate of the backward reaction, so the concentrations of reactants and products stay constant.

- **Dynamic:** both reactions are still going on all the time. Molecules keep changing into products and back again; it is only the overall amounts that stop changing.
- **Constant, not equal:** the concentrations of reactants and products are steady, but not usually equal to each other.
- **Closed system:** nothing can enter or leave. If a product gas escapes from an open flask, the backward reaction can't happen and equilibrium is never reached.
- **Either direction:** the same equilibrium mixture is reached whether you start from the reactants or from the products (under the same conditions).

The **position of equilibrium** describes the make-up of the equilibrium mixture. If it "lies to the right", the mixture contains mostly products; if it "lies to the left", mostly reactants. A change that increases the yield of product is said to shift the position of equilibrium to the right.

Le Chatelier's principle

KEY IDEA

Le Chatelier's principle: if a system at equilibrium is subjected to a change in conditions, the position of equilibrium shifts in the direction that tends to oppose (counteract) the change.

The system can never cancel the change completely, only partly. You use the principle to predict qualitatively which way the position of equilibrium moves.

Changing concentration

Adding more of a reactant shifts the equilibrium to the **right**, using up some of the added reactant and making more product. Removing a product also shifts it to the right, replacing some of what was removed. Iodine in water shows this with a colour change:

$$I_2(aq) + 2OH^-(aq) \rightleftharpoons I^-(aq) + IO^-(aq) + H_2O(l)$$

(iodine solution is brown; the ions on the right are colourless)

Adding sodium hydroxide increases $[OH^-]$, so the equilibrium shifts right to remove hydroxide ions: the brown colour fades. Adding acid removes OH^- ($H^+ + OH^- \rightarrow H_2O$), so the equilibrium shifts back to the left and the brown colour returns.

Changing pressure

Pressure only matters for reactions involving gases. Increasing the pressure shifts the equilibrium towards the side with **fewer gas molecules**, because fewer molecules exert less pressure, which opposes the change. Decreasing the pressure shifts it towards the side with more gas molecules. If both sides have the same number of gas molecules, as in $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, changing the pressure has **no effect** on the position of equilibrium (although it does change the concentrations of everything).

EXAM TIP

A model answer for "What is the effect of increasing the pressure on the yield of methanol, $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$?": The equilibrium shifts to oppose the increase in pressure, towards the side with fewer gas molecules. There are 3 moles of gas on the left and only 1 on the right, so the position of equilibrium moves to the right and the yield of methanol increases.

Changing temperature

Raising the temperature shifts the equilibrium in the **endothermic** direction, which takes in heat and so opposes the rise. Lowering the temperature shifts it in the **exothermic** direction, which gives out heat. The ΔH quoted with an equation is for the forward reaction; the backward reaction has the same value with the opposite sign.

EXAM TIP

A model answer for "What is the effect of increasing the temperature on the yield of ammonia, $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta H = -92$ kJ mol^{-1}?": The equilibrium shifts to oppose the increase in temperature, in the endothermic direction. The forward reaction is exothermic, so the backward reaction is endothermic: the position of equilibrium moves to the left and the yield of ammonia decreases.

Adding a catalyst

A catalyst has **no effect on the position of equilibrium**. It lowers the activation energy for the forward and the backward reactions by the same amount, so it speeds up both by the same factor. The equilibrium mixture is the same; the system just reaches it sooner.

Equilibrium in industry: compromise conditions

A chemical manufacturer wants a high **yield**, but also a fast **rate** and a low **cost**. These pull in different directions, so industrial conditions are a compromise. Most of the important industrial equilibria have an exothermic forward reaction and fewer gas molecules on the right, so the same arguments come up again and again.

- **Temperature:** a low temperature would give a high equilibrium yield (the forward reaction is exothermic), but the rate would be far too slow. A moderate temperature is chosen as a compromise between yield and rate.

- **Pressure:** a high pressure increases both the equilibrium yield (fewer gas molecules on the right) and the rate (gases are more concentrated). But very high pressures need a lot of electrical energy to run the compressors, and thick-walled, expensive equipment that is more dangerous if it fails. So the pressure is no higher than it is worth paying for.
- **Catalyst:** always used. It doesn't change the yield, but it allows a lower temperature to be used while still giving a good rate.
- **Recycling:** the reactants that have not reacted are separated from the products and fed back into the reactor, so very little is wasted even if the yield in each pass is low.

Process	Equation	Typical conditions	Notes
Haber process (ammonia)	$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ $\Delta H = -92$ kJ mol^{-1}	about 450 °C, about 200 atm, iron catalyst	Ammonia is removed by cooling it until it liquefies; unreacted gases are recycled.
Contact process (sulfuric acid)	$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $\Delta H = -197$ kJ mol^{-1}	about 450 °C, 1–2 atm, vanadium(V) oxide catalyst	The equilibrium already lies far to the right, so high pressure would add cost for very little extra yield.
Methanol	$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$ $\Delta H = -91$ kJ mol^{-1}	about 250 °C, 50–100 atm, copper-based catalyst	Moderate temperature and pressure for the same yield–rate–cost reasons.
Ethanol from ethene	$C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$ $\Delta H = -46$ kJ mol^{-1}	about 300 °C, 60–70 atm, phosphoric acid catalyst on a silica support	Only about 5% of the ethene is converted in each pass, but recycling raises the overall conversion to about 95%. Too high a pressure would also make the ethene polymerise.

Quick checks

QUICK CHECK

What does the total area under a Maxwell–Boltzmann distribution curve represent?

- The activation energy
- The total number of molecules
- The mean energy of the molecules
- The number of molecules that can react

The curve shows how many molecules have each energy, so adding them all up (the area) gives the total number of molecules. Only the area beyond E_a is the number that can react.

QUICK CHECK

Which statement best explains why a small increase in temperature greatly increases the rate of a reaction?

- The activation energy decreases.
- The particles collide more often.
- A much greater proportion of the colliding particles have energy at least equal to the activation energy.
- The enthalpy change of the reaction becomes more negative.

The tail of the Maxwell–Boltzmann curve beyond E_a grows a lot when the temperature rises. More frequent collisions also help, but that effect is small. The activation energy itself does not change with temperature.

QUICK CHECK

How does a catalyst increase the rate of a reaction?

- It gives the reacting particles more kinetic energy.
- It increases the frequency of collisions between particles.
- It makes the reaction more exothermic.
- It provides an alternative route with a lower activation energy.

With a lower E_a, more of the existing molecules have enough energy to react. The distribution of energies and ΔH are unchanged.

QUICK CHECK

For $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $\Delta H = -92$ kJ mol^{-1}, what happens when the temperature is increased?

- The rate increases and the equilibrium yield of ammonia decreases.
- The rate increases and the equilibrium yield of ammonia increases.
- The rate decreases and the equilibrium yield of ammonia increases.
- The rate increases and the equilibrium yield of ammonia stays the same.

Higher temperature always increases the rate. The equilibrium shifts in the endothermic direction, which is backwards here because the forward reaction is exothermic, so less ammonia is present at equilibrium.

QUICK CHECK

The pressure on an equilibrium mixture of $H_2(g)$, $I_2(g)$ and HI(g) is doubled by halving its volume. What do you observe?

- The purple colour fades as the equilibrium shifts to the right.
- The purple colour fades as the equilibrium shifts to the left.
- The purple colour darkens, and the position of equilibrium does not change.
- There is no change in colour.

There are two gas molecules on each side, so pressure does not move the position of equilibrium. But the same amount of iodine is now in half the volume, so its concentration doubles and the colour darkens.

QUICK CHECK

Which statement about a system in dynamic equilibrium is correct?

- The concentrations of the reactants and products are equal.
- The forward and backward reactions continue at equal rates.
- Both the forward and backward reactions have stopped.
- Adding a catalyst moves the position of equilibrium to the right.

At dynamic equilibrium both reactions keep going at the same rate, so the concentrations stay constant, but they are not usually equal. A catalyst speeds up both reactions equally and does not move the position of equilibrium.

QUICK CHECK

Why is the Contact process run at only 1–2 atm even though higher pressure would increase the yield of SO_3?

- Higher pressure would slow the reaction down.
- The catalyst only works at low pressure.
- Higher pressure would shift the equilibrium to the left.
- The yield is already very high, so the extra cost of high pressure is not worth it.

$2SO_2 + O_2 \rightleftharpoons 2SO_3$ has fewer gas molecules on the right, so higher pressure would increase the yield, but the equilibrium already lies far to the right at moderate pressure. Compressors and strong equipment would cost more than the small gain is worth.

AS · Unit 2 · WCH12 Energetics, Group Chemistry, Halogenoalkanes and Alcohols

Halogenoalkanes, alcohols and spectra

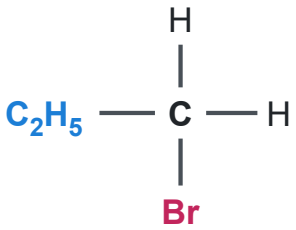
Swap one hydrogen in an alkane for a halogen or an –OH group and the molecule suddenly has a polar bond that other species can attack. This topic follows what happens next: how halogenoalkanes are substituted or lose HX to give alkenes, how alcohols are oxidised, halogenated and dehydrated, how the products are separated and purified in the lab, and how mass spectra and infrared spectra tell you which molecule you have made.

Halogenoalkanes: names and classes

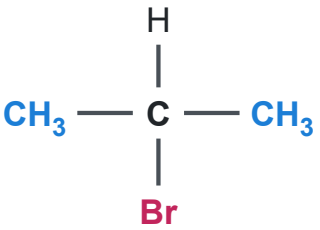
A halogenoalkane is an alkane in which one or more hydrogen atoms have been replaced by halogen atoms. The name is built on the longest carbon chain, with a prefix for each halogen: **fluoro-**, **chloro-**, **bromo-** or **iodo-**. Number the chain from the end that gives the substituents the lowest numbers, and write the prefixes in **alphabetical order** (bromo before chloro before methyl). Multiplying prefixes such as di- and tri- are ignored when you put the prefixes in order.

- **$CH_3CH_2CH_2Br$:** 1-bromopropane
- **$CH_3CHClCH_3$:** 2-chloropropane
- **$(CH_3)_3CBr$:** 2-bromo-2-methylpropane
- **$CH_3CHBrCHClCH_3$:** 2-bromo-3-chlorobutane (b comes before c, and numbering from either end gives 2 and 3, so bromo, first alphabetically, takes the lower number)
- **CH_2BrCH_2Br:** 1,2-dibromoethane

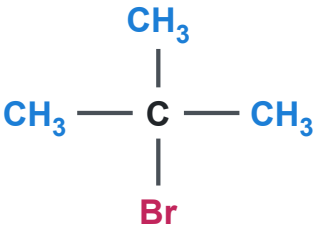
Halogenoalkanes are classified by looking at the carbon atom that carries the halogen and counting how many **other carbon atoms** are bonded to it. This matters because, as you will see, the class changes both the mechanism and the speed of the reactions.



1-bromopropane
primary
1 carbon attached



2-bromopropane
secondary
2 carbons attached



2-bromo-2-methylpropane
tertiary
3 carbons attached

Count the carbon groups (blue) on the carbon that carries the halogen: one for primary, two for secondary, three for tertiary.

Halogenoalkanes are polar molecules but they cannot form hydrogen bonds with water, so they do not mix with it. Their boiling temperatures rise with chain length and from chloro to bromo to iodo, because bigger molecules with more electrons have stronger London forces between them. Branched isomers boil lower than straight-chain ones because their molecules touch over a smaller area.

Why halogenoalkanes react: the polar C–X bond

Every halogen is more electronegative than carbon, so the bonding pair in the C–X bond is pulled towards the halogen. The carbon becomes slightly positive (δ+) and the halogen slightly negative (δ−). That electron-poor carbon is a target for **nucleophiles**.

DEFINITION

A **nucleophile** is a species that donates a lone pair of electrons to form a new covalent bond. Important nucleophiles in this topic are the hydroxide ion, $:OH^-$, the cyanide ion, $:CN^-$, ammonia, $:NH_3$, and water, $H_2O:$.

In a **nucleophilic substitution** the nucleophile replaces the halogen, which leaves as a halide ion, X^-. The C–X bond breaks **heterolytically**: both electrons of the bond go to the halogen.

KEY IDEA

A **curly arrow** shows the movement of a **pair of electrons**. It always starts from a lone pair or from the middle of a bond, and its head points to where the pair ends up: an atom (forming a new bond) or an atom that leaves with the pair (breaking a bond).

Bond strength decides the rate

You might expect the most polar bond, C–F, to react fastest because its carbon is the most positive. Experiments show the opposite. The step that controls the rate is breaking the C–X bond, and the bonds get weaker down the group: as the halogen atom gets bigger, the shared pair of electrons is further from its nucleus and is held less strongly.

Bond	Mean bond enthalpy / kJ mol^{-1}	Rate of substitution
C–F	467	so slow that fluoroalkanes are effectively unreactive
C–Cl	346	slow
C–Br	290	faster

C–I	228	fastest

Nucleophilic substitution reactions

The same pattern works for every nucleophile: the lone pair attacks the δ+ carbon, the C–X bond breaks and the halide ion leaves. What changes is the product, and the solvent and conditions chosen to make that product.

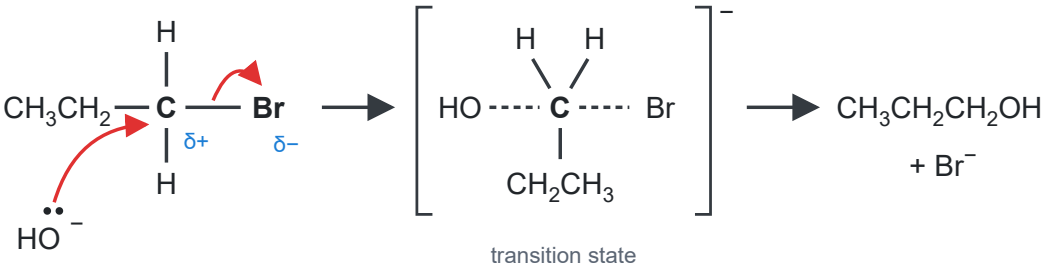
Reagent and conditions	Role of the reagent	Product from 1-bromopropane
KOH or NaOH dissolved in **water**, heat under reflux	nucleophile, OH^-	propan-1-ol, $CH_3CH_2CH_2OH$
water, with aqueous silver nitrate and ethanol, warm	nucleophile, H_2O (hydrolysis)	propan-1-ol, and a precipitate of AgBr
KCN dissolved in **ethanol**, heat under reflux	nucleophile, CN^-	butanenitrile, $CH_3CH_2CH_2CN$
excess concentrated NH_3 in **ethanol**, heat in a sealed tube	nucleophile, NH_3	propylamine (propan-1-amine), $CH_3CH_2CH_2NH_2$

With aqueous hydroxide ions: making alcohols

Heating a halogenoalkane under reflux with aqueous potassium or sodium hydroxide replaces the halogen with –OH:

$CH_3CH_2CH_2Br + OH^- \rightarrow CH_3CH_2CH_2OH + Br^-$ or $CH_3CH_2CH_2Br + KOH \rightarrow CH_3CH_2CH_2OH + KBr$

The hydroxide ion carries a full negative charge, so it is attracted far more strongly to the δ+ carbon than a neutral water molecule is. That is why OH^- substitutes much faster than water does. For a primary halogenoalkane the whole change happens in **one step**:



The lone pair on oxygen attacks the δ+ carbon from the side opposite the bromine. As the new O–C bond forms, the C–Br bond breaks and both its electrons go with the bromide ion.

EXAM TIP

For full marks in a mechanism: show the δ+ and δ− on the C–Br bond, a lone pair on the nucleophile, an arrow from that lone pair to the carbon, an arrow from the C–Br bond to the bromine, and the products including the halide ion. Make sure the first arrow starts at the lone pair, not at the minus sign.

With cyanide ions: making nitriles

Warming a halogenoalkane under reflux with potassium cyanide dissolved in ethanol gives a **nitrile**. The cyanide ion attacks through the lone pair on its carbon atom, so the reaction adds one carbon to the chain, which makes it useful in synthesis:

$CH_3CH_2CH_2Br + CN^- \rightarrow CH_3CH_2CH_2CN + Br^-$ (1-bromopropane gives butanenitrile: four carbons, counting the one in –C≡N)

The mechanism is the same as with hydroxide: an arrow from the lone pair on the carbon of $:C{\equiv}N^-$ to the δ+ carbon, and an arrow from the C–Br bond to Br. Potassium cyanide is extremely toxic, so it is handled in a fume cupboard and never acidified (that would release HCN gas).

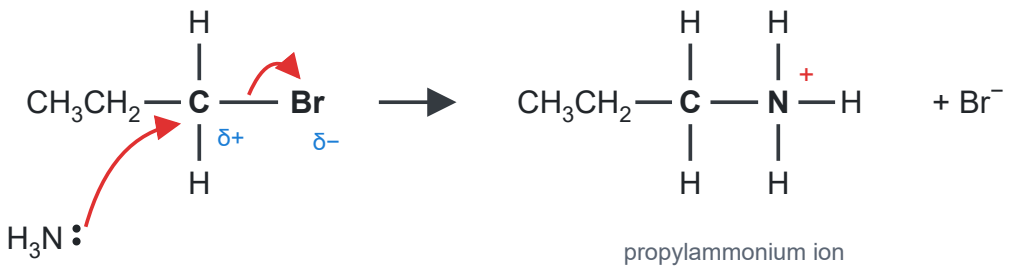
With ammonia: making amines

Ammonia's lone pair on nitrogen lets it act as a nucleophile too. The halogenoalkane is heated with an **excess of concentrated ammonia in ethanol** in a **sealed tube**. Ammonia is a gas that would simply boil out of an open flask, so sealing the tube keeps its concentration high; the pressure inside rises as it is heated.

$CH_3CH_2CH_2Br + 2NH_3 \rightarrow CH_3CH_2CH_2NH_2 + NH_4Br$

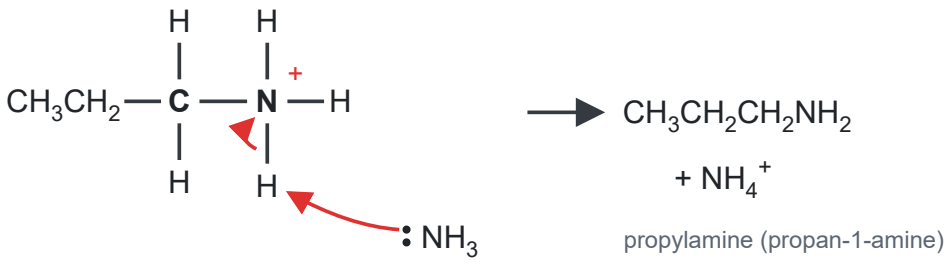
Mechanism: 1-bromopropane with ammonia

1. The lone pair on the nitrogen of NH_3 attacks the δ+ carbon, and the C–Br bond breaks heterolytically. Nitrogen now has four bonds and no lone pair, so it carries a positive charge: the product of this step is the propylammonium ion, $CH_3CH_2CH_2NH_3^+$, plus Br^-.



propylammonium ion

2. A second ammonia molecule acts as a base. Its lone pair removes H^+ from the N–H bond of the ammonium ion, and the electrons of that N–H bond return to nitrogen as a lone pair.



The ammonium ion and the bromide ion end up together as ammonium bromide, NH_4Br, which is why two moles of ammonia appear in the overall equation.

The amine product still has a lone pair on nitrogen, so it is a nucleophile as well. It can attack another halogenoalkane molecule to form a secondary amine, then a tertiary amine, and finally a quaternary ammonium salt. Using a **large excess of ammonia** makes it far more likely that a halogenoalkane molecule meets an ammonia molecule than an amine molecule, which keeps these further substitutions, and the loss of yield they cause, to a minimum.

Comparing rates of hydrolysis

Hydrolysis means splitting a molecule by reaction with water. Water is a weak nucleophile, but warm halogenoalkanes do react with it slowly:

$$R–X + H_2O \rightarrow R–OH + H^+ + X^-$$

If silver nitrate solution is present, each halide ion released is caught straight away as an insoluble silver halide, $Ag^+(aq) + X^-(aq) \rightarrow AgX(s)$. The silver ions do not attack the halogenoalkane themselves; they just reveal how fast the halide ions are being set free. So **the sooner a precipitate appears, the faster the hydrolysis**. Ethanol is added as a solvent that mixes with both the halogenoalkane and the aqueous silver nitrate, so the reactants are in one phase.

CORE PRACTICAL

Investigating the rates of hydrolysis of halogenoalkanes

Put equal volumes of ethanol and aqueous silver nitrate into a set of test tubes and stand them in a water bath at about 50 °C. Add the same number of drops of each halogenoalkane at the same moment and time how long each takes to form a precipitate.

- **Changing the halogen** (1-chlorobutane, 1-bromobutane, 1-iodobutane): the iodo compound gives a yellow precipitate of AgI first, then the bromo compound gives cream AgBr, and the chloro compound gives white AgCl last. This follows the C–X bond enthalpies: C–I is the weakest bond and breaks most easily.
- **Changing the structure** (1-bromobutane, 2-bromobutane, 2-bromo-2-methylpropane): the tertiary compound reacts fastest and the primary slowest, because tertiary halogenoalkanes react through a stable carbocation (the S_N1 route below).
- **Fair test:** keep the temperature, the volumes and the amount of each halogenoalkane the same, and change only one feature at a time. When comparing halogens, keep the carbon skeleton the same.
- **Safety:** ethanol and many halogenoalkanes are flammable, so heat with a water bath and no naked flame. Silver nitrate stains skin; wear gloves and eye protection.

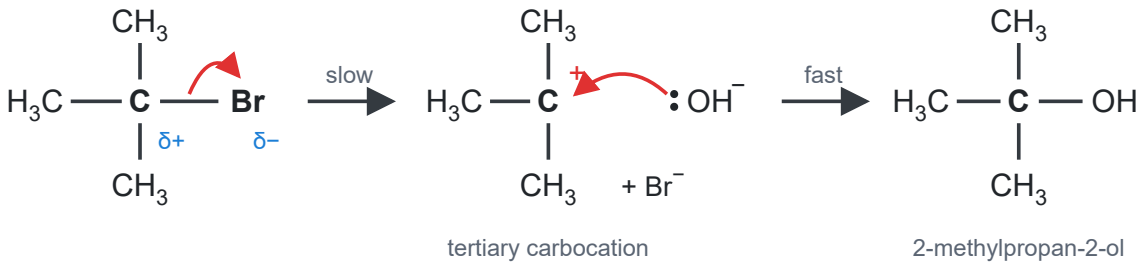
COMMON MISTAKE

Don't explain the halogen order using polarity or electronegativity. C–F is the *most* polar bond, yet fluoroalkanes are the *slowest* to react. The deciding factor is bond enthalpy: the weaker the C–X bond, the faster it breaks.

S_N2 and S_N1: primary versus tertiary

Nucleophilic substitution can happen in two different ways, and which one a halogenoalkane uses depends on its class. The names say what happens: **S**ubstitution, **N**ucleophilic, and the number of species involved in the slowest (rate-determining) step.

- **S_N2 (primary halogenoalkanes):** one step. The nucleophile attacks as the C–X bond breaks, passing through a transition state containing both the halogenoalkane and the nucleophile. The rate depends on both concentrations: rate = k[RX][OH^-].
- **S_N1 (tertiary halogenoalkanes):** two steps. First the C–X bond breaks on its own to give a carbocation and a halide ion (slow). Then the nucleophile attacks the carbocation (fast). Only the halogenoalkane is in the slow step: rate = k[RX].
- **Secondary halogenoalkanes** can react by either route, and often use a mixture of both.



S_N1 for a tertiary halogenoalkane. The C–Br bond breaks first (slow) to leave a planar carbocation; the hydroxide ion then attacks it (fast).

Why does structure decide the route? Two effects work together.

- **Carbocation stability:** alkyl groups push electron density towards a neighbouring positive carbon (a positive inductive effect), spreading out its charge. A tertiary carbocation has three alkyl groups doing this and is stable enough to form; a primary carbocation has only one and is too unstable, so primary halogenoalkanes cannot take the S_N1 route.
- **Crowding:** in S_N2 the nucleophile must reach the carbon from the side opposite the halogen. Three bulky methyl groups on a tertiary carbon block that approach, while the small hydrogen atoms on a primary carbon leave it open.

KEY IDEA

Tertiary halogenoalkanes hydrolyse fastest because their route, through a stabilised tertiary carbocation, has a lower activation energy than the routes open to secondary and primary halogenoalkanes. That is why the order of reactivity in the silver nitrate experiment is tertiary > secondary > primary.

Elimination: making alkenes

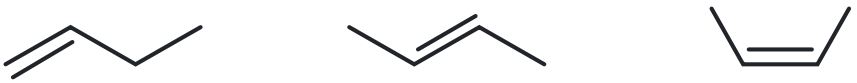
Change the solvent and the same reagent does something different. Heated under reflux with potassium hydroxide dissolved in **ethanol**, a halogenoalkane loses a hydrogen halide and forms an alkene. Here the hydroxide ion acts as a **base**: it removes an H^+ from a carbon *next to* the C–X carbon, a C=C double bond forms between those two carbons, and the halide ion leaves.

$CH_3CH_2CH_2Br + KOH \rightarrow CH_3CH{=}CH_2 + KBr + H_2O$

	KOH in water	KOH in ethanol
Role of OH^-	nucleophile	base

Type of reaction	nucleophilic substitution	elimination
Organic product	alcohol	alkene
Favoured by	primary halogenoalkanes, lower temperature	tertiary halogenoalkanes, higher temperature

In practice both reactions compete and you usually get a mixture; the conditions just tip the balance. If the halogen is not on the end of an unsymmetrical chain, the hydrogen can be removed from a carbon on either side, so more than one alkene forms. 2-Bromobutane gives three products, because but-2-ene has E and Z isomers:



but-1-ene (E)-but-2-ene (Z)-but-2-ene

Removing H from carbon 1 gives but-1-ene; removing it from carbon 3 gives but-2-ene, which exists as E and Z isomers.

Alcohols: names, classes and shape

Alcohols contain the hydroxyl group, –OH, and the open-chain ones with one –OH have the general formula $C_nH_{2n+1}OH$. Their names end in **-ol**, with a number showing where the –OH is: butan-2-ol is $CH_3CH(OH)CH_2CH_3$. With two or three –OH groups, keep the "e" of the alkane name and add **-diol** or **-triol**: ethane-1,2-diol ($HOCH_2CH_2OH$) and propane-1,2,3-triol (glycerol). When the molecule has a group that takes the suffix, such as a carboxylic acid, the –OH becomes the prefix **hydroxy-**: $CH_3CH(OH)COOH$ is 2-hydroxypropanoic acid.

Alcohols are classified exactly like halogenoalkanes, by counting the carbon atoms bonded to the carbon that carries the –OH:

- **Primary:** propan-1-ol, $CH_3CH_2CH_2OH$ (one carbon attached; methanol, with none, behaves like a primary alcohol)
- **Secondary:** propan-2-ol, $CH_3CH(OH)CH_3$ (two carbons attached)
- **Tertiary:** 2-methylpropan-2-ol, $(CH_3)_3COH$ (three carbons attached)

Around each carbon there are four bonding pairs, so the bonds are arranged tetrahedrally at about 109.5°. The oxygen has two bonding pairs and two lone pairs. Lone pairs repel more strongly than bonding pairs, so they squeeze the C–O–H angle down to about **104.5°**, giving a bent shape like water.

The O–H group lets alcohol molecules hydrogen bond to one another, so alcohols boil far higher than alkanes of similar mass, and the small ones mix completely with water. Branching lowers the boiling temperature, as for

halogenoalkanes: butan-1-ol boils at 118 °C but its more compact isomer 2-methylpropan-2-ol boils at 82 °C.

Reactions of alcohols

Combustion

Alcohols burn in plenty of air with a clean, almost colourless flame to give carbon dioxide and water, which is why ethanol is used as a fuel:

$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$

Halogenation: swapping –OH for a halogen

The –OH group can be replaced by a halogen, turning the alcohol into a halogenoalkane. Different reagents are used for each halogen.

Halogen	Reagent	Equation (ethanol)
Cl	phosphorus(V) chloride, PCl_5, at room temperature	$C_2H_5OH + PCl_5 \rightarrow C_2H_5Cl + POCl_3 + HCl$
Cl (tertiary alcohols)	concentrated hydrochloric acid, shaken at room temperature	$(CH_3)_3COH + HCl \rightarrow (CH_3)_3CCl + H_2O$
Br	potassium bromide and 50% concentrated sulfuric acid, which make HBr in the flask	$KBr + H_2SO_4 \rightarrow KHSO_4 + HBr$ $C_2H_5OH + HBr \rightarrow C_2H_5Br + H_2O$
I	red phosphorus and iodine, warmed, which make phosphorus(III) iodide in the flask	$2P + 3I_2 \rightarrow 2PI_3$ $3C_2H_5OH + PI_3 \rightarrow 3C_2H_5I + H_3PO_3$

The reaction with PCl_5 gives off **steamy fumes of hydrogen chloride**, which turn damp blue litmus red and form white smoke with ammonia. This is a test for an –OH group. It only works on a dry sample, because water and carboxylic acids contain –OH and give the same fumes.

COMMON MISTAKE

Why not make HBr or HI from the salt and fully concentrated sulfuric acid? Concentrated H_2SO_4 is an oxidising agent: it oxidises some bromide to bromine and almost all iodide to iodine (Topic 8). Diluting it to about 50% reduces the oxidation for bromide, and for iodide a different route through PI_3 is used.

Oxidation with acidified potassium dichromate(VI)

Potassium dichromate(VI), $K_2Cr_2O_7$, acidified with dilute sulfuric acid oxidises primary and secondary alcohols. As it does so, the **orange** dichromate(VI) ion, $Cr_2O_7^{2-}$, is reduced to the **green** chromium(III) ion, Cr^{3+}. Equations use [O] to stand for oxygen supplied by the oxidising agent.

Oxidation removes two hydrogen atoms: the one on the –OH and one from the carbon carrying the –OH. A C=O bond forms in their place. This is why the class of alcohol matters so much.

Alcohol	Conditions	Product	Equation
primary	limited oxidising agent (alcohol in excess); warm gently and **distil off** the product as it forms	aldehyde	$CH_3CH_2CH_2OH + [O] \rightarrow CH_3CH_2CHO + H_2O$
primary	excess oxidising agent; **heat under reflux**	carboxylic acid	$CH_3CH_2CH_2OH + 2[O] \rightarrow CH_3CH_2COOH + H_2O$
secondary	heat under reflux	ketone	$CH_3CH(OH)CH_3 + [O] \rightarrow CH_3COCH_3 + H_2O$
tertiary	heat under reflux	no reaction: the solution stays orange	—

A **tertiary** alcohol has no hydrogen atom on the carbon bonded to the –OH, so a C=O bond cannot form without breaking a C–C bond, and dichromate cannot do that. A **ketone** is not oxidised further for the same reason. An **aldehyde** still has a hydrogen on its C=O carbon, so it is easily oxidised on to a carboxylic acid.

So how do you stop at the aldehyde? Use boiling temperatures. Propanal (49 °C) has no O–H group, so its molecules cannot hydrogen bond to each other, and it boils far lower than propan-1-ol (97 °C) or propanoic acid (141 °C). Gentle warming in a distillation apparatus lets the aldehyde escape the flask as soon as it forms, before it meets more oxidising agent. Heating under reflux does the opposite: everything is returned to the flask until it has all been oxidised to the acid.

KEY IDEA

Aldehyde names end in **-al** and the C=O is always on carbon 1, so no number is needed (propanal, CH_3CH_2CHO; write CHO, never COH). Ketone names end in **-one** with a number when the chain is long enough to need one (propanone, but pentan-2-one and pentan-3-one).

Telling the products apart

Test	Positive result	What it shows

warm with Fehling's or Benedict's solution	blue solution gives a red precipitate of copper(I) oxide, Cu_2O	aldehyde (ketones give no change)
add sodium hydrogencarbonate or sodium carbonate	effervescence; the gas turns limewater cloudy	carboxylic acid
warm with acidified $K_2Cr_2O_7$	orange to green	primary or secondary alcohol (or aldehyde)
add PCl_5 to the dry sample	steamy fumes of HCl	–OH group (alcohol or carboxylic acid)

The Fehling's and Benedict's tests work because an aldehyde can be oxidised and a ketone cannot. The Cu^{2+} ions in the alkaline blue solution oxidise the aldehyde and are themselves reduced to copper(I):

$CH_3CHO + 2Cu^{2+} + 5OH^- \rightarrow CH_3COO^- + Cu_2O + 3H_2O$

Dehydration: making alkenes

Heating an alcohol with **concentrated phosphoric(V) acid**, H_3PO_4, removes a molecule of water and gives an alkene. The acid is a catalyst and dehydrating agent, and the reaction is an **acid-catalysed elimination**. The alkene, with its much lower boiling temperature, can be distilled off as it forms.

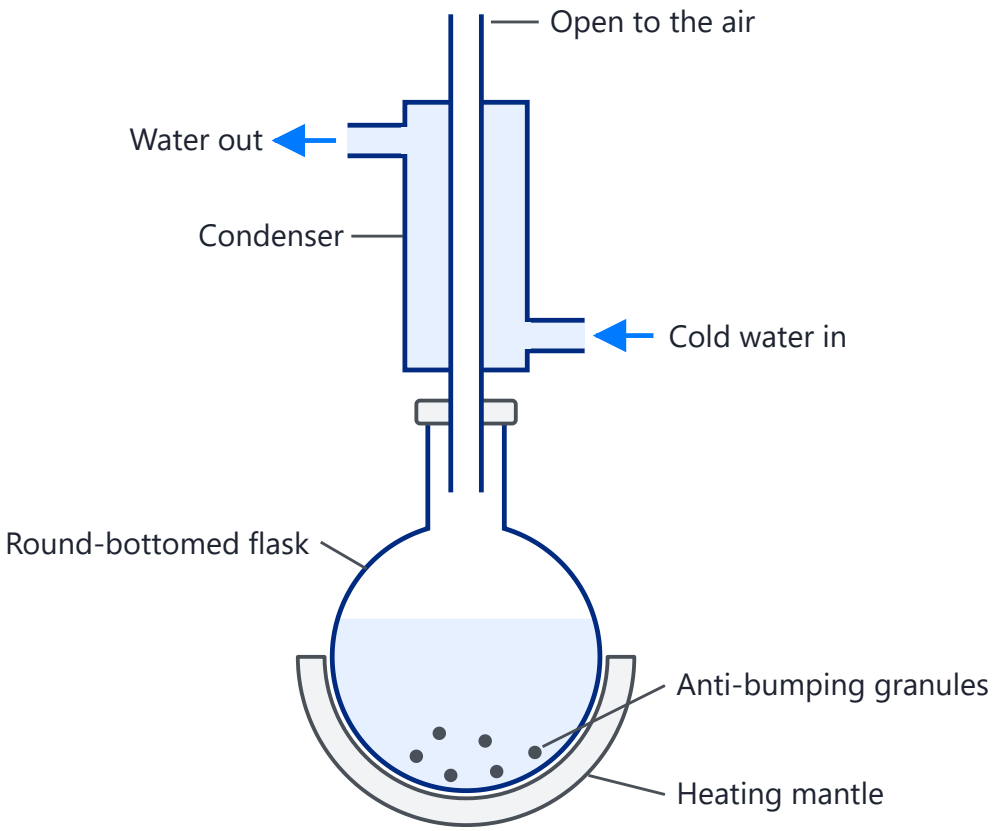
$CH_3CH_2CH_2OH \rightarrow CH_3CH{=}CH_2 + H_2O$

As with halogenoalkanes, an –OH group in the middle of an unsymmetrical chain can lose a hydrogen from either neighbouring carbon. Butan-2-ol gives but-1-ene and both E- and Z-but-2-ene. Making alkenes this way from ethanol produced by fermentation gives a route to polymers such as poly(ethene) that does not depend on crude oil.

Practical techniques for organic liquids

Heating under reflux

Many organic reactions are slow and their reactants are volatile and flammable. Heating under **reflux** lets you heat a mixture for a long time without losing anything: a vertical condenser turns the vapours back into liquid, which drips back into the flask.

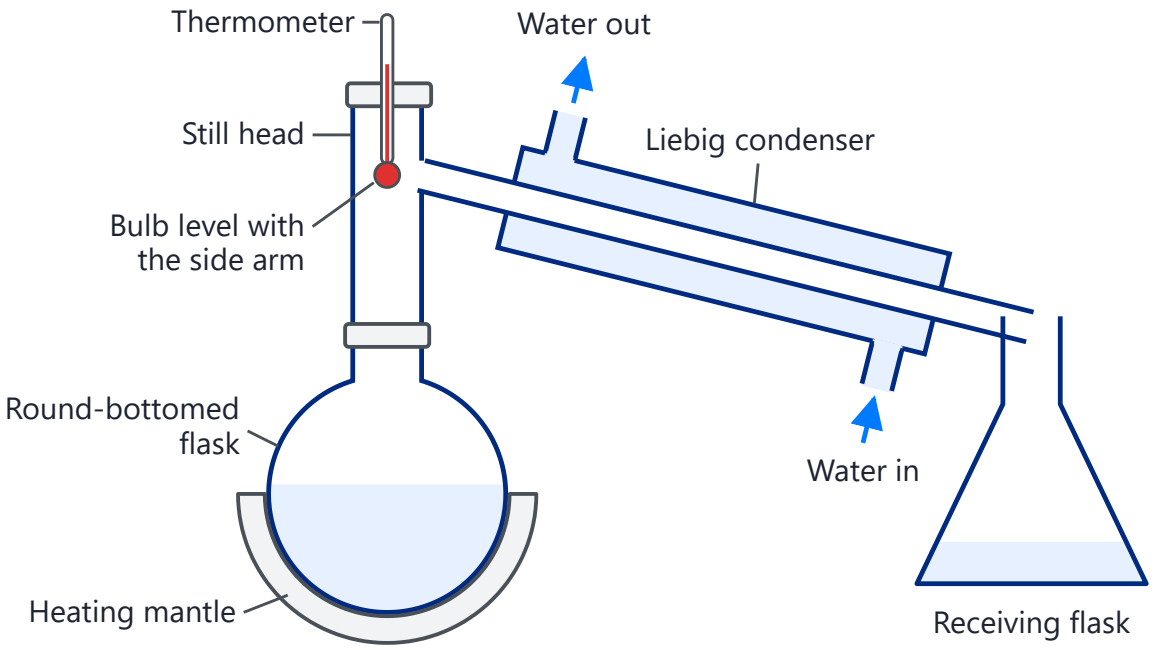


Heating under reflux. Water enters the condenser jacket at the bottom and leaves at the top, and the top of the condenser stays open.

- **Never seal the top of the condenser.** Heating a closed system builds up pressure until the apparatus bursts.
- **Anti-bumping granules** give the liquid many small places to form bubbles, so it boils smoothly instead of in sudden violent bursts.
- **Heat electrically** (a heating mantle or water bath), not with a Bunsen flame, because most organic liquids catch fire easily.

Distillation

Distillation separates a liquid from a mixture by boiling it off and condensing the vapour somewhere else. It is used to remove a product as it forms (the aldehyde above) and to purify a liquid at the end.



Simple distillation. The thermometer bulb sits opposite the side arm so it reads the temperature of the vapour that is distilling over.

- The thermometer bulb must be level with the side arm, so it measures the vapour leaving the flask, not the liquid boiling below.
- Collect only the fraction that distils within a degree or two of the product's boiling temperature.
- Water enters the condenser at the lower end so the jacket stays full of cold water.
- When drawing apparatus, don't draw lines across the joints between flask, still head and condenser (that would mean they were sealed off), and draw the condenser's water jacket as a closed outer tube with two open water connections.

Separating, washing and drying

- **Separating funnel:** two liquids that don't mix form two layers. Run off the lower, denser layer through the tap and keep whichever layer holds your product. If you are not sure which layer is which, add a little water and see which one gets bigger.
- **Washing:** shake the organic layer with sodium hydrogencarbonate solution to neutralise any acid left in it. CO_2 forms, so invert the funnel and open the tap often to release the pressure. Washing with water or saturated sodium chloride solution removes water-soluble impurities.
- **Drying:** add an anhydrous salt such as anhydrous sodium sulfate, magnesium sulfate or calcium chloride, swirl, and leave until the liquid is clear and the solid no longer clumps. A drying agent must absorb water without dissolving in or reacting with the product. Then decant or filter the liquid off.
- **Solvent extraction:** shaking a mixture with a solvent that dissolves the product better (iodine moves from water into cyclohexane, for example) concentrates it in one layer, which can then be separated and distilled.

- **Boiling temperature:** a pure liquid boils at a fixed, known temperature, so measuring it is a check on purity and identity. It is not proof on its own, since different liquids can share a boiling temperature, and it changes with atmospheric pressure.

CORE PRACTICAL

Chlorination of 2-methylpropan-2-ol with concentrated hydrochloric acid

Shake 2-methylpropan-2-ol with concentrated hydrochloric acid in a separating funnel for several minutes, releasing the pressure often. Tertiary alcohols react quickly at room temperature, and 2-chloro-2-methylpropane forms as an upper layer that does not mix with the aqueous acid.

- Run off and discard the lower aqueous layer.
- Shake the organic layer with sodium hydrogencarbonate solution to remove leftover acid, releasing the CO_2 pressure, and discard the aqueous layer again.
- Dry the product with anhydrous sodium sulfate, decant it into a clean flask and distil, collecting the fraction that boils at about 50–52 °C.
- **Safety:** concentrated HCl is corrosive and gives off choking fumes, so work in a fume cupboard with eye protection. The product is very flammable.

CORE PRACTICAL

Oxidation of propan-1-ol to propanal and propanoic acid

To make propanal: warm propan-1-ol gently in a distillation apparatus and add the acidified dichromate(VI) a drop at a time, so that the alcohol is always in excess and the propanal distils off as soon as it forms. Keep the receiver cool, since propanal boils at only 49 °C.

To make propanoic acid: heat propan-1-ol with an excess of acidified dichromate(VI) under reflux for about 20 minutes, then rearrange the apparatus for distillation and distil off the acid.

In both cases the mixture turns from orange to green. Test the products: propanal gives a red precipitate with Fehling's solution; propanoic acid fizzes with sodium hydrogencarbonate.

Mass spectra of organic compounds

In a mass spectrometer, a sample molecule M is hit by high-energy electrons, which knock one electron out of it. That leaves a positive ion with an unpaired electron, the **molecular ion**:

$M(g) \rightarrow [M]^{+\bullet}(g) + e^-$ for example $CH_3CH_2CH_2OH \rightarrow [CH_3CH_2CH_2OH]^{+\bullet} + e^-$

The dot shows the unpaired electron and is often left out. The ion has a charge of +1, so its m/z value equals its mass. The **molecular ion peak**, normally the peak with the highest m/z, therefore gives the **relative molecular mass** of the compound.

Fragmentation

Many molecular ions have so much excess energy that a covalent bond breaks. One piece keeps the positive charge and the other becomes an uncharged free radical:

$[M]^{+\bullet} \rightarrow X^+ + Y^\bullet$

Only charged particles are deflected and detected, so the spectrum shows X^+ but never $Y^\bullet$. Fragment ions that are relatively stable, such as carbocations stabilised by alkyl groups and acylium ions $[RCO]^+$, form most often and give the tallest peaks. The tallest peak of all is called the **base peak**.

m/z	Common fragment ion
15	CH_3^+
29	$C_2H_5^+$ or CHO^+
31	CH_2OH^+ (typical of primary alcohols)
43	$C_3H_7^+$ or CH_3CO^+
45	$COOH^+$, or CH_3CHOH^+ from some secondary alcohols
57	$C_4H_9^+$ or $C_2H_5CO^+$

Structural isomers have the same molecular ion but break up in different places, so their fragments tell them apart. Pentan-2-one and pentan-3-one both have $M_r = 86$. Only pentan-2-one has a CH_3–C=O group, so only it gives a large peak at m/z = 43:

$[CH_3COCH_2CH_2CH_3]^{+\bullet} \rightarrow [CH_3CO]^+ + {}^\bullet CH_2CH_2CH_3 \quad (m/z = 43)$

Pentan-3-one has an ethyl group on each side of the C=O, so it gives $[CH_3CH_2CO]^+$ at m/z = 57 and $CH_3CH_2^+$ at m/z = 29 instead:

$[CH_3CH_2COCH_2CH_3]^{+\bullet} \rightarrow [CH_3CH_2CO]^+ + {}^\bullet CH_2CH_3 \quad (m/z = 57)$

EXAM TIP

When asked to identify a peak, give the formula of the ion **with its positive charge**, for example $[CH_3CO]^+$. A fragmentation equation must show the ion on one side and a radical (uncharged) on the other.

Infrared spectroscopy

Covalent bonds are not rigid: they constantly stretch and bend. Each kind of bond vibrates at its own natural frequency, which depends on the masses of the atoms and the strength of the bond. When infrared radiation of exactly that frequency shines on the molecule, the bond absorbs it and vibrates more strongly. An infrared spectrometer passes a range of IR frequencies through a sample and records how much of each gets through. Dips (troughs) in the transmittance show which bonds are present.

Positions are given as **wavenumbers**, the number of waves per centimetre, $\tilde{\nu} = 1/\lambda$, in cm^{-1}. A bigger wavenumber means a higher frequency. Strong bonds and bonds to light hydrogen atoms absorb at high wavenumbers.

KEY IDEA

A vibration absorbs IR only if it **changes the polarity** (dipole) of the molecule. C=O, O–H, N–H and C–H bonds all do. A molecule made of two identical atoms, such as N_2, O_2 or H_2, has no dipole however it stretches, so it does not absorb IR at all.

Bond	Where it occurs	Wavenumber / cm^{-1}	Appearance
O–H	alcohols	3200–3600	broad, because of hydrogen bonding
O–H	carboxylic acids	2500–3300	very broad, often overlapping the C–H peaks
N–H	amines	3300–3500	medium, sharper than O–H
C–H	almost all organic compounds	2850–3100	sharp
C≡N	nitriles	2200–2260	sharp, medium
C=O	aldehydes, ketones, carboxylic acids, esters	1680–1750	strong and sharp
C=C	alkenes	1620–1680	weak to medium
C–O	alcohols, acids, esters	1000–1300	strong, in the fingerprint region

These ranges are approximate; in the exam, always quote the range from the data booklet you are given. Above about 1500 cm^{-1} the troughs are few and easy to assign, so this is the region for identifying functional groups. Below 1500 cm^{-1} lies the **fingerprint region**: a complicated pattern of bending and skeletal vibrations that is unique to each compound. A computer can match it against a database of known spectra to identify the substance exactly.

EXAM TIP

Following a reaction by IR: oxidising propan-1-ol to propanal makes the broad O–H trough at 3200–3600 cm^{-1} disappear and a strong C=O trough appear near 1730 cm^{-1}. Going on to propanoic acid brings back a very broad O–H trough, now at 2500–3300 cm^{-1}, alongside the C=O.

Infrared absorption and the greenhouse effect

The Earth's surface, warmed by sunlight, radiates energy back out as infrared. Bonds in molecules of carbon dioxide, water vapour and methane absorb some of this IR in exactly the way described above, then re-emit it in all directions, some back towards the ground. This traps energy in the lower atmosphere, so these are **greenhouse gases**. Nitrogen and oxygen, which make up 99% of dry air, cannot absorb IR because their vibrations never change their polarity. Carbon dioxide is non-polar overall, but its asymmetric stretching and bending vibrations do create a temporary dipole, so it absorbs strongly.

Quick checks

QUICK CHECK

What is the name of $CH_3CHClCBr(CH_3)CH_3$?

- 3-chloro-2-bromo-2-methylbutane
- 2-bromo-3-chloro-2-methylbutane
- 2-methyl-2-bromo-3-chlorobutane
- 3-bromo-2-chloro-3-methylbutane

The longest chain has four carbons (butane). Numbering from the end nearest the Br gives the substituents positions 2, 2 and 3 rather than 2, 3 and 3. The prefixes then go in alphabetical order: bromo, chloro, methyl. Putting chloro first ignores the alphabetical rule.

QUICK CHECK

1-Chlorobutane, 1-bromobutane and 1-iodobutane are warmed with aqueous silver nitrate in ethanol. Which statement explains the order in which the precipitates appear?

- The C–Cl bond is the most polar, so 1-chlorobutane reacts first.
- Iodide ions are the best nucleophiles, so the iodo compound reacts first.
- Silver ions attack the C–I bond fastest because iodine is the largest atom.
- The C–I bond has the lowest bond enthalpy, so 1-iodobutane is hydrolysed fastest.

Water is the nucleophile, and breaking the C–X bond controls the rate. C–I (about 228 kJ mol^{-1}) is much weaker than C–Cl (346 kJ mol^{-1}), so the yellow AgI appears first. Polarity predicts the wrong order: C–Cl is more polar than C–I, yet slower.

QUICK CHECK

The rate equation for the hydrolysis of a halogenoalkane by hydroxide ions is rate = k[RBr]. Which halogenoalkane is it most likely to be?

- 1-bromobutane
- bromomethane
- 2-bromo-2-methylpropane
- 1-bromo-2-methylpropane

A rate that does not depend on $[OH^-]$ means hydroxide is not in the slow step: the S_N1 route, where the C–Br bond breaks first to give a carbocation. Only the tertiary compound forms a stable enough carbocation. 1-Bromo-2-methylpropane is branched, but the carbon carrying the Br has only one other carbon on it, so it is primary.

QUICK CHECK

2-Bromopropane is heated under reflux with potassium hydroxide dissolved in ethanol. What is the main organic product and what is the role of the hydroxide ion?

- propan-2-ol; nucleophile
- propene; base
- propene; nucleophile
- propan-2-ol; base

In ethanol the hydroxide ion removes an H^+ from the carbon next to the C–Br carbon, so it acts as a base and an alkene forms by elimination. In water it would act as a nucleophile and give the alcohol.

QUICK CHECK

Which alcohol does *not* change the colour of acidified potassium dichromate(VI) when heated with it?

- butan-1-ol
- butan-2-ol
- 2-methylpropan-1-ol
- 2-methylpropan-2-ol

2-Methylpropan-2-ol is tertiary: the carbon carrying the –OH has no hydrogen, so no C=O can form and the dichromate stays orange. 2-Methylpropan-1-ol is branched but primary (its –OH carbon has two hydrogens), so it is oxidised.

QUICK CHECK

Why is propanal distilled off as it forms when propan-1-ol is oxidised?

- So it leaves the oxidising agent before it can be oxidised further to propanoic acid
- Because propanal decomposes if it is heated under reflux
- Because distillation speeds up the oxidation of the alcohol
- So the dichromate(VI) can be reused

An aldehyde is easily oxidised to a carboxylic acid. Propanal boils at 49 °C, well below the alcohol and the acid, because its molecules cannot hydrogen bond to each other, so it can be removed from the hot mixture as soon as it is made.

QUICK CHECK

An IR spectrum has a strong, sharp absorption at 1715 cm^{-1} and no absorption between 2500 and 3600 cm^{-1} apart from sharp peaks near 2950 cm^{-1}. The compound has M_r = 58. What is it?

- propan-1-ol
- propanoic acid
- propanone
- prop-2-en-1-ol

The C=O trough at 1715 cm^{-1} rules out the alcohols, and the absence of any broad O–H trough rules out the acid (and the alcohols). Propanone, CH_3COCH_3, has M_r = 58. Prop-2-en-1-ol has the same M_r but would show a broad alcohol O–H trough and no C=O.

QUICK CHECK

Which gas in the atmosphere does *not* absorb infrared radiation?

- carbon dioxide
- methane
- water vapour
- nitrogen

A bond absorbs IR only if its vibration changes the polarity of the molecule. N≡N joins two identical atoms, so stretching it never creates a dipole. CO_2 is non-polar overall, but its asymmetric stretch and bending vibrations do produce a dipole, so it absorbs.